

Measurement of the neutrino magnetic moment at the NOvA experiment

Technical note

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Abstract

This is the abstract

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1 Introduction

In this analysis, we aim to detect a potential signal of the effective muon neutrino magnetic moment in the NuMI Off-axis ν_e Appearance (NOvA) Near Detector (ND). This signal would manifest as an excess of neutrino-on-electron (ν -on- e) elastic scattering interactions at low electron recoil energies, proportional to the value of the effective neutrino magnetic moment, over the Standard Model (SM) background. If no significant excess is observed, we will establish an upper limit on the effective muon neutrino magnetic moment.

Detecting the neutrino magnetic moment (μ_ν) would provide definitive evidence of new Beyond Standard Model (BSM) physics, and measuring its value would help identify the appropriate BSM theory. As current and planned experiments can only detect an anomalously large neutrino magnetic moment, observing such a signature would strongly suggest that neutrinos are Majorana particles and would have significant implications for astrophysics and cosmology [1].

The best model-independent experimental results on the neutrino magnetic moment come from experiments searching for dark matter using xenon-based detectors. These highly sensitive detectors detect solar neutrinos, which are part of the background in dark matter searches but can be reanalysed for other purposes. In 2020, the XENON1T experiment observed [2] a low energy excess of solar neutrinos, which could correspond to a signal from an anomalously large effective magnetic moment within $\mu_{\nu_\odot} \in (0.14, 0.29) \times 10^{-10} \mu_B$ at 90% Confidence Level (C.L.), where $\nu_\odot$ marks solar neutrinos. However, this result was disfavoured by the follow-up XENONnT experiment in 2022 [3], which saw no excess and set the current world-leading limit on neutrino magnetic moment at $\mu_{\nu_\odot} < 0.063 \times 10^{-10} \mu_B$ at 90% C.L.. Other solar neutrino experiments also reported null results regarding neutrino magnetic moment [4, 5], placing less stringent limits on its value. Given some basic assumptions [5, 6] this limit for solar neutrinos would correspond to a limit on muon neutrino effective magnetic moment of $\mu_{\nu_\mu} < 0.137 \times 10^{-10} \mu_B$. However, the relationship between effective magnetic moments of different neutrino flavours may be non-trivial, especially in the context of possible new BSM physics, and studying muon neutrinos remains an important endeavour [7].

The best results for ν_μ and $\bar{\nu}_\mu$ come from accelerator-based stopped pion neutrino sources [8, 9], which also do not observe any low energy excess and provide an upper limit on the effective muon neutrino magnetic moment of $\mu_{\nu_\mu} < 6.8 \times 10^{-10} \mu_B$ at 90% C.L. [8]. Stopped pion neutrino sources provide well-understood beams made up of ν_μ , $\bar{\nu}_\mu$ and ν_e with energies up to 52.8 MeV. Slightly looser limits come from pion decay-in-flight accelerator-based measurements (similar to NOvA) [10, 11], which provide a limit of $\mu_{\nu_\mu} < 8.5 \times 10^{-10} \mu_B$ at 90% C.L..

Thanks to the very intense and highly pure beam of muon neutrinos and antineutrinos, and a detector designed for the reconstruction and identification of events with electrons in the final state, NOvA is well-positioned to provide a highly competitive, and possibly even world-leading, measurement (or limit) of the effective muon neutrino magnetic moment. A previous analysis of NOvA ND data for a measurement of the effective muon neutrino magnetic moment was presented in a thesis [12], providing a (statistics-only) limit of $\mu_{\nu_\mu} < 15.8 \times 10^{-10} \mu_B$ at 90% C.L..

Additionally, ν -on- e elastic scattering interactions are used in various other analyses in NOvA, specifically in efforts to constrain the neutrino beam prediction [13] and in the search for Light Dark Matter (LDM) [14]. These analyses developed various tools and methods that can be utilized in the search for a neutrino magnetic moment.

(TO DO: Possibly include cosmological implications and limits)

In this technical note, we will provide an overview of the theory of neutrino electromagnetic interactions in Sec. 2, focusing on the effective neutrino magnetic moment and its implications for ν -on- e

measurements and other theoretical considerations. In Sec. 3, we will discuss the analysis strategy, the signal and background definition, as well as the data and simulation samples and the analysis weights. Following this, in Sec. 4 we will explain the selection of events for this analysis, while in Sec. 5 we will address the relevant systematic uncertainties and in Sec. 6 the statistical methods employed to analyse the results. We will present the results of this analysis in Sec. 7. Finally, section 8 will summarise the findings of this analysis.

2 Theoretical Overview

Neutrinos in the SM are massless and electrically neutral particles. However, even SM neutrinos can have electromagnetic interaction through loop diagrams involving charged leptons and the W boson, covered by the neutrino charge radius [1].

In general BSM theories, considering interactions with a single photon as shown on Fig. 1, neutrino electromagnetic interactions can be described by an effective interaction Hamiltonian [15]

$$\mathcal{H}_{em}^{(v)}(x) = \sum_{k,j=1}^N \bar{\nu}_k(x) \Lambda_{\mu}^{kj} \nu_j(x) A^{\mu}(x). \quad (1)$$

Here $\nu_k(x)$, $k = 1, \dots, N$, are neutrino fields in the mass basis with N neutrino mass states, Λ_{μ}^{kj} is a general vertex function and $A^{\mu}(x)$ is the electromagnetic field.

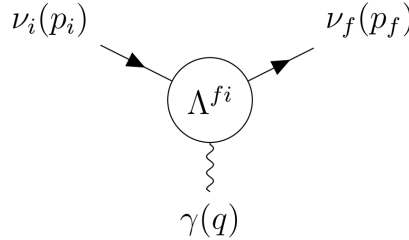


Figure 1: Effective coupling of neutrinos with one photon electromagnetic field.

The vertex function $\Lambda_{\mu}^{fi}(q)$ is generally a matrix and, in the most general case consistent with the SM gauge invariance [16, 17], can be written in terms of linearly independent products of Dirac matrices (γ) and only depends on the four momentum of the photon ($q = p_f - p_i$):

$$\Lambda_{\mu}^{fi}(q) = \mathbb{F}_1^{fi}(q^2) q_{\mu} + \mathbb{F}_2^{fi}(q^2) q_{\mu} \gamma_5 + \mathbb{F}_3^{fi}(q^2) \gamma_{\mu} + \mathbb{F}_4^{fi}(q^2) \gamma_{\mu} \gamma_5 + \mathbb{F}_5^{fi}(q^2) \sigma_{\mu\nu} q^{\nu} + \mathbb{F}_6^{fi}(q^2) \varepsilon_{\mu\nu\rho\gamma} q^{\nu} \sigma^{\rho\gamma}, \quad (2)$$

where $\mathbb{F}_i^{fi}(q^2)$ are six Lorentz invariant form factors and δ and ε are the Dirac delta and the Levi-Civita symbols respectively.

Applying conditions of hermiticity ($\mathcal{H}_{em}^{(v)\dagger} = \mathcal{H}_{em}^{(v)}$) and of the gauge invariance of the electromagnetic field, the vertex function can be rewritten as

$$\Lambda_{\mu}^{fi}(q) = (\gamma_{\mu} - q_{\mu} \not{q} / q^2) \left[\mathbb{F}_Q^{fi}(q^2) + \mathbb{F}_A^{fi}(q^2) q^2 \gamma_5 \right] - i \sigma_{\mu\nu} q^{\nu} \left[\mathbb{F}_M^{fi}(q^2) + i \mathbb{F}_E^{fi}(q^2) \gamma_5 \right], \quad (3)$$

where $\mathbb{F}_Q^{fi}$, $\mathbb{F}_M^{fi}$, $\mathbb{F}_E^{fi}$ and $\mathbb{F}_A^{fi}$ are hermitian matrices representing the charge, dipole magnetic, dipole electric and anapole neutrino form factors respectively. It is clear that the vertex function only depends on the

square of the four momentum of the photon q^2 . In coupling with a real photon ($q^2 = 0$) these form factors become the neutrino charge and magnetic, electric and anapole moments respectively. Additionally, the neutrino charge radius corresponds to the second term in the expansion of the charge form factor [15].

The above expression can be simplified [18] as

$$\Lambda_\mu^{fi}(q) = \gamma_\mu \left(Q_{\nu_{fi}} + \frac{q^2}{6} \langle r^2 \rangle_{\nu_{fi}} \right) - i \sigma_{\mu\nu} q^\nu \mu_{\nu_{fi}}, \quad (4)$$

where $Q_{\nu_{fi}}$, $\langle r^2 \rangle_{\nu_{fi}}$, and $\mu_{\nu_{fi}}$ are the neutrino charge, effective charge radius (also containing anapole moment), and an effective magnetic moment (also containing electric moment) respectively. This is possible thanks to the similar effects of the neutrino charge radius and the anapole moment, and of the neutrino magnetic and electric moments, on neutrino interactions. Therefore, these are the three neutrino electromagnetic properties (charge, effective charge radius and effective magnetic moment) measured in experiments.

The neutrino electric charge is primarily constrained through measurements of the neutrality of matter and through cosmological observations, which provide much better constraints than neutrino oscillation experiments [15]. On the other hand, the neutrino charge radius would manifest as an increase in the size of the ν -on- e elastic scattering coupling constants, allowing it to be studied in neutrino oscillation experiments such as NOvA. Additionally, the value of the neutrino charge radius in the SM is only an order of magnitude smaller than the current world-leading limits [19] and measuring it could either confirm the validity of neutrino interactions in the SM, or open possibilities to non-standard contributions to neutrino scattering [15]. However, measurement of the neutrino charge radius is not part of this analysis, but may be included in the future re-analysis of the ν -on- e interactions in the NOvA ND.

2.1 Neutrino Electric and Magnetic Dipole Moments

The size and effect of neutrino electromagnetic properties depend on the specific BSM theory applied. Evaluating one loop diagrams in the minimally extended SM with three right-handed Dirac neutrinos gives the first approximation of the electric and magnetic moments, which are now 3×3 matrices with elements:

$$\left. \begin{matrix} \mu_{kj}^D \\ i\epsilon_{kj}^D \end{matrix} \right\} \simeq \frac{3eG_F}{16\sqrt{2}\pi^2} (m_k \pm m_j) \left(\delta_{kj} - \frac{1}{2} \sum_{l=e,\mu,\tau} U_{lk}^* U_{lj} \frac{m_l^2}{m_W^2} \right), \quad (5)$$

where m_k, m_j are the neutrino masses and m_l are the masses of charged leptons which appear in the loop diagrams [15]. The D superscript denotes Dirac neutrinos and M denotes Majorana neutrinos throughout this section. Also, e is the electron charge, G_F is the Fermi coupling constant, U is the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) neutrino oscillation matrix, and m_W is the mass of the W boson. Higher order electromagnetic corrections were neglected, but can also have a significant contribution, depending on the theory.

It can be seen that Dirac neutrinos have no diagonal electric moments ($\epsilon_{kk}^D = 0$) and their diagonal magnetic moments are approximately

$$\mu_{kk}^D \simeq \frac{3eG_F m_k}{8\sqrt{2}\pi^2} \simeq 3.2 \times 10^{-19} \left(\frac{m_k}{\text{eV}} \right) \mu_B, \quad (6)$$

where μ_B is the Bohr magneton which represents the value of the electron magnetic moment [15]. Neutrino magnetic moments are therefore strongly suppressed by the smallness of neutrino masses, with theoretical predictions in Eq. 6 several orders of magnitude below the reach of current experiments [18].

The transition magnetic moments in the minimally extended SM from Eq. 5 are suppressed with respect to the largest of the diagonal magnetic moments by at least a factor of 10^{-4} due to the m_W^2 in the denominator. The transition electric moments are even smaller due to the mass difference in Eq. 5. Therefore an experimental observation of a magnetic moment larger than in Eq. 6 would indicate physics beyond the minimally extended SM [15, 20].

The suppression of the neutrino magnetic moment by the smallness of its mass can be also expressed in a general case [20]. The ‘natural’ upper limits on the size of the neutrino magnetic moment for any BSM theory that has New Physics (NP) generated at a scale Λ_{NP} can be expressed as [21]

$$\mu_V^D(\mu_B) \lesssim 3 \times 10^{-15} \frac{m_V^D(\text{eV})}{[\Lambda_{NP}(\text{TeV})]^2}. \quad (7)$$

Therefore for $\Lambda_{NP} \simeq 1 \text{ TeV}$ and $m_V^D \lesssim 1 \text{ eV}$ the limit becomes $\mu_V^D \lesssim 3 \times 10^{-15} \mu_B$, well below the current experimental capabilities. However, these upper bounds only apply if NP is generated well above the electroweak scale $\Lambda_{EW} \sim 100 \text{ GeV}$ [15].

For Majorana neutrinos, the magnetic and electric form factors (and therefore the magnetic and electric moment matrices) are antisymmetric, thus Majorana neutrinos only have transition moments. The simplest extension of the SM that includes Majorana neutrinos requires either the addition of a Higgs triplet, or right-handed neutrinos together with a Higgs singlet [15]. Neglecting the Feynman diagrams which depend on the model of the scalar sector, the magnetic and electric dipole moments are

$$\left. \begin{matrix} \mu_{kj}^M \\ \epsilon_{kj}^D \end{matrix} \right\} \simeq \mp \frac{3ieG_F}{16\sqrt{2}\pi^2} (m_k \pm m_j) \sum_{l=e,\mu,\tau} \text{Im/Re} [U_{lk}^* U_{lj}] \frac{m_l^2}{m_W^2}, \quad (8)$$

where Im is for μ_{kj}^M and Re is for ϵ_{kj}^D . These are difficult to compare to the Dirac case, due to possible presence of Majorana phases in the PMNS matrices, but it is clear that they have the same order of magnitude as Dirac transition dipole moments. However, the neglected model dependent contributions can enhance the transition dipole moments for Majorana neutrinos [15].

The natural upper bound on the Majorana magnetic moment is less strict compared to the Dirac neutrinos, due to the antisymmetric nature of Majorana magnetic moment, which requires additional Yukawa couplings in the BSM theory compared to Dirac neutrinos, which can enhance the maximal possible magnetic moment [20]. The limit for Majorana neutrinos can be expressed as

$$\mu_{\alpha\beta}^M(\mu_B) \leq 4 \times 10^{-9} \frac{[m_V^M]_{\alpha\beta}(\text{eV})}{[\Lambda_{NP}(\text{TeV})]^2} \frac{m_\tau^2}{|m_\alpha^2 - m_\beta^2|}, \quad \alpha, \beta \in \{e, \mu, \tau\}. \quad (9)$$

Here, the neutrino magnetic moment is expressed in the flavour basis instead of the mass basis, since the charged lepton masses are diagonal here. The two bases are related by

$$\mu_{ij} = \sum_{\alpha\beta} \mu_{\alpha\beta} U_{\alpha i}^* U_{\beta j}. \quad (10)$$

and the effect of the neutrino magnetic moment on neutrino interactions does not depend on the choice of the basis[22].

These considerations imply, that if a magnetic moment $\mu \gtrsim 10^{-15} \mu_B$ were measured, neutrinos are almost certainly Majorana particles [20].

2.1.1 Effective Neutrino Magnetic Moment

As mentioned above, the neutrino magnetic moment measured in experiments is the so-called effective neutrino magnetic moment, which is a combination of electric and magnetic dipole moments and depends on the neutrino source and oscillations. In the ultra-relativistic limit, the effective neutrino magnetic moment is

$$\mu_{\nu_l}^2(L, E_\nu) = \sum_j \left| \sum_k U_{lk}^* e^{\mp i \Delta m_{kj}^2 L / 2E_\nu} (\mu_{jk} - i \varepsilon_{jk}) \right|^2, \quad (11)$$

where the minus sign in the exponent is for neutrinos and the plus sign for antineutrinos [15]. Therefore, the only difference between the effective neutrino and antineutrino magnetic moment is in the phase induced by neutrino oscillations. For experiments with baselines short enough that neutrino oscillations would not have time to develop ($\Delta m^2 L / 2E_\nu \ll \sim 1$), such as the NOvA ND, the effective magnetic moment is the same for neutrinos and antineutrinos and is independent of the neutrino energy.

Since the effective magnetic moment depends on the initial neutrino flavour, it is different for experiments studying neutrinos from different sources. Additionally, experiments such as solar neutrino experiments, need to include matter effects on the neutrino oscillations. Therefore the reports on the value (or upper limit) of the effective neutrino magnetic moment are not directly comparable between different types of neutrino experiments.

2.2 Measuring the Neutrino Magnetic Moment

The most sensitive method to measure the neutrino magnetic moment is the low energy elastic scattering of (anti)neutrinos on electrons [15]. The schematic diagram for this interaction is shown in Fig. 2, where the recoil electron's kinetic energy is defined as ($T_e = E_{e'} - m_e$) and the recoil angle with respect to the incoming neutrino beam (θ) is shown.

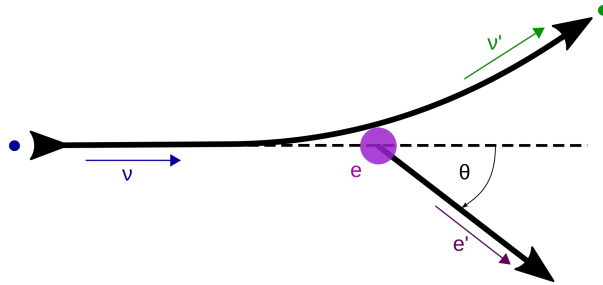


Figure 2: Neutrino-on-electron elastic scattering diagram

Since the ν -on- e interaction is governed by simple $2 \rightarrow 2$ kinematics, it can be shown that

$$(P_\nu - P_{e'})^2 = (P_{\nu'} - P_e)^2, \quad (12)$$

$$m_\nu^2 + m_e^2 - 2E_\nu E_{e'} + 2E_\nu p_{e'} \cos \theta = m_\nu^2 + m_e^2 - 2E_{\nu'} m_e. \quad (13)$$

From the energy conservation

$$E_\nu + m_e = E_{\nu'} + E_{e'} = E_{\nu'} + T_e + m_e \Rightarrow E_{\nu'} = E_\nu - T_e \quad (14)$$

it follows that

$$E_\nu p_{e'} \cos \theta = E_\nu E_{e'} - E_{\nu'} m_e = E_\nu (T_e + m_e) - (E_\nu - T_e) m_e = T_e (E_\nu + m_e), \quad (15)$$

$$\cos \theta = \frac{E_\nu + m_e}{E_\nu} \sqrt{\frac{T_e^2}{E_\nu^2 - m_e^2}} = \frac{E_\nu + m_e}{E_\nu} \sqrt{\frac{T_e^2}{T_e^2 + 2T_e m_e}}. \quad (16)$$

And finally

$$\cos \theta = \frac{E_\nu + m_e}{E_\nu} \sqrt{\frac{T_e}{T_e + 2m_e}}. \quad (17)$$

Which can be rearranged to get

$$T_e = \frac{2m_e E_\nu^2 \cos^2 \theta}{(E_\nu + m_e)^2 - E_\nu^2 \cos^2 \theta}. \quad (18)$$

The electron's kinetic energy is therefore constrained as

$$T_e \leq \frac{2E_\nu^2}{2E_\nu + m_e}, \quad (19)$$

160 which corresponds to the limit $\cos \theta \rightarrow 1$ when the recoil electron goes exactly forward in the incident
161 neutrino direction, as depicted in Fig. 3.

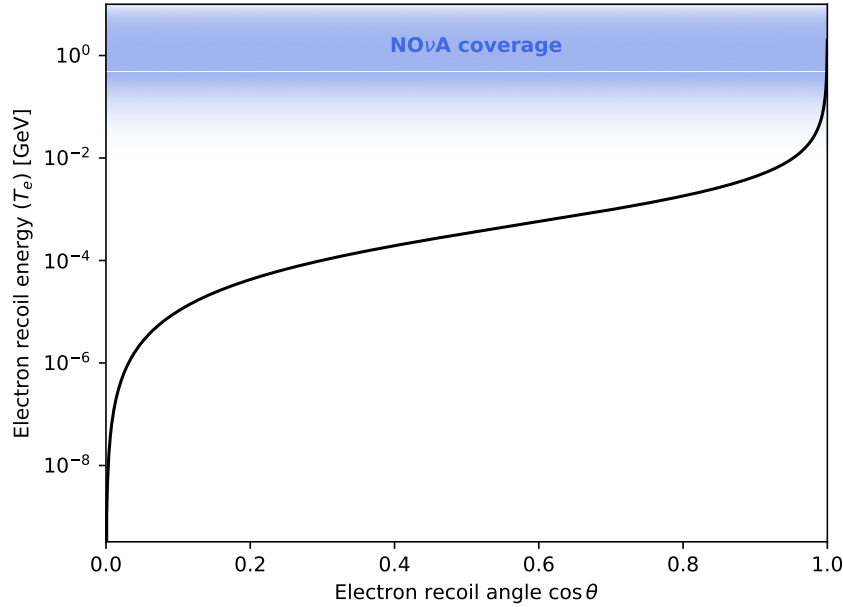


Figure 3: Relation between the recoil electron's kinetic energy and angle for the ν -on- e elastic scattering. The coverage of the NOvA detectors for measuring the electron recoil energy is shown in blue. Only very forward electrons are therefore recorded in NOvA.

Considering $E_\nu \sim \text{GeV}$, it is useful to approximate $\frac{m_e^2}{E_\nu^2} \rightarrow 0$. Additionally, considering only very small electron recoil angles, meaning $\theta^2 \cong (1 - \cos^2 \theta)$, applied to Eq. 17 results in

$$T_e \theta^2 \cong T_e \left(1 - \left(\frac{E_\nu + m_e}{E_\nu} \right)^2 \frac{T_e}{T_e + 2m_e} \right) = T_e \left(1 - \left(1 + \frac{2m_e}{E_\nu} \right) \frac{T_e}{T_e + 2m_e} \right), \quad (20)$$

therefore

$$T_e \theta^2 \cong \frac{2m_e T_e}{T_e + 2m_e} \left(1 - \frac{T_e}{E_v}\right) = 2m_e \left(\frac{1}{1 + \frac{2m_e}{T_e}}\right) \left(1 - \frac{T_e}{E_v}\right), \quad (21)$$

and finally

$$T_e \theta^2 \cong 2m_e \left(1 - \frac{T_e}{E_v}\right) < 2m_e. \quad (22)$$

162 This is a strong limit that very clearly distinguishes the ν -on- e elastic scattering events from other
163 similar interactions involving single electron (mainly the ν_e Charged Current (CC) interactions).

164 2.2.1 Neutrino Magnetic Moment Cross Section

In the ultra-relativistic limit, the neutrino magnetic moment interaction flips the neutrino helicity, while the SM weak interaction conserves it, which means it is possible to add the two contributions to the total ν -on- e cross section incoherently (without interference terms) [15]:

$$\frac{d\sigma_{\nu\text{-on-}e}}{dT_e} = \left(\frac{d\sigma_{\nu\text{-on-}e}}{dT_e}\right)_{\text{SM}} + \left(\frac{d\sigma_{\nu\text{-on-}e}}{dT_e}\right)_{\text{MAG}}. \quad (23)$$

The SM contribution can be expressed as [23, 15]:

$$\left(\frac{d\sigma_{\nu\text{-on-}e}}{dT_e}\right)_{\text{SM}} = \frac{2G_F^2 m_e}{\pi} \left\{ g_1^2 + g_2^2 \left(1 - \frac{T_e}{E_v}\right)^2 - g_1 g_2 \frac{m_e T_e}{E_v^2} \right\}, \quad (24)$$

where the coupling constants g_1 and g_2 differ between neutrino flavours and between neutrinos and antineutrinos. Their values are:

$$g_1^{\nu_e} = g_2^{\bar{\nu}_e} = \sin^2 \theta_W + 1/2, \quad g_2^{\nu_e} = g_1^{\bar{\nu}_e} = \sin^2 \theta_W, \quad (25)$$

$$g_1^{\nu_{\mu,\tau}} = g_2^{\bar{\nu}_{\mu,\tau}} = \sin^2 \theta_W - 1/2, \quad g_2^{\nu_{\mu,\tau}} = g_1^{\bar{\nu}_{\mu,\tau}} = \sin^2 \theta_W, \quad (26)$$

165 where $\sin^2 \theta_W \cong 0.23$.

166 The total SM cross section, and therefore the number of SM ν -on- e interactions, depends on the
167 neutrino energy and the minimum measured electron recoil energy. However, in general the cross section
168 for ν_e is about 2.5 times larger than for the $\bar{\nu}_e$, about 6 times larger than for $\nu_{\mu/\tau}$ and about 7 times
169 larger than for $\bar{\nu}_{\mu/\tau}$.

The neutrino magnetic moment contribution is [24, 15]:

$$\left(\frac{d\sigma_{\nu\text{-on-}e}}{dT_e}\right)_{\text{MAG}} = \frac{\pi \alpha^2}{m_e^2} \left(\frac{1}{T_e} - \frac{1}{E_v}\right) \left(\frac{\mu_{\nu_l}}{\mu_B}\right)^2, \quad (27)$$

170 where α is the fine structure constant and μ_{ν_l} is the effective magnetic moment of ν_l . The total cross
171 section now only depends on the neutrino energy and on the effective magnetic moment, but is the same
172 for neutrinos and antineutrinos.

173 The comparison of the SM and the neutrino magnetic moment differential cross sections is shown in
174 Fig.4. Whereas the SM cross section is approximately uniform for $T_e \rightarrow 0$, the neutrino magnetic moment
175 cross section rises to infinity. However, this reach is limited by the experimental capabilities of detecting
176 electrons with very low energies. The (possible) NOvA coverage is shown with a shaded blue region,

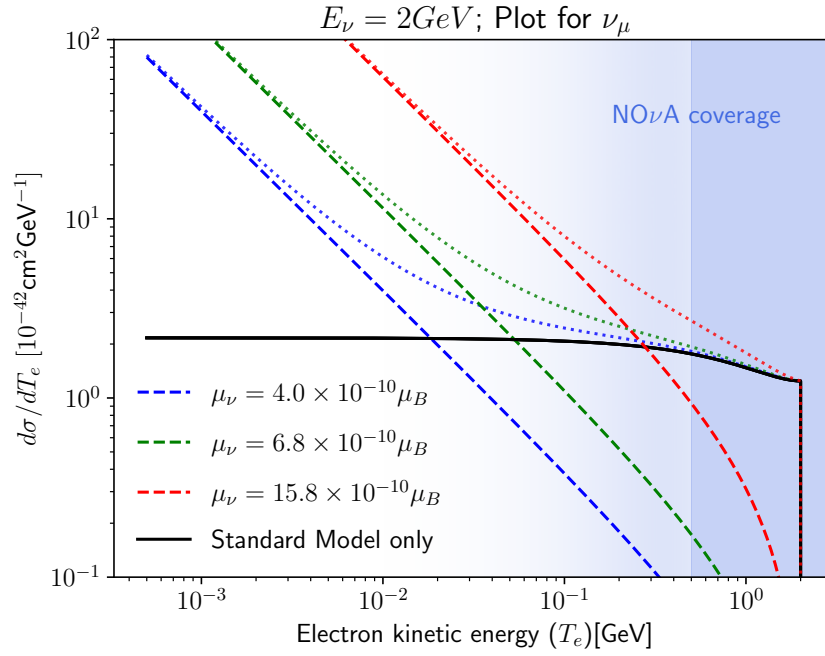


Figure 4: Comparison of the neutrino magnetic moment (coloured) and the SM (black) cross sections for the ν -on- e elastic scattering. Different colours depict different values of the neutrino magnetic moment, with red corresponding to the previous NO ν A measurement, green the LSND result, and blue a possible ultimate NO ν A sensitivity, as discussed in the introduction to this chapter. Dashed lines are the individual cross sections and dotted lines are the added total cross section with the standard model contribution. NO ν A coverage of electron recoil energies is shown in shaded blue.

with current capability reaching $T_e = 0.5 \text{ GeV}$. Future analyses might extend this reach to lower T_e , with the lowest possible detectable electron recoil energy $T_{e,min} \approx 0.01 \text{ GeV}$.

Calculating the ratio of the neutrino magnetic moment and the SM cross sections, as shown in Fig. 5, can serve as a proxy to estimate the number of neutrino magnetic moment events in relation to the predicted number of SM events, if the E_ν and T_e are known. Additionally, comparing the ratio of the total cross sections can reveal the expected total number of neutrino magnetic moment events as a function of the predicted number of SM events. Considering $E_\nu = 2 \text{ GeV}$, $\mu_\nu = 6.8 \times 10^{-10} \mu_B$ (current best limit for ν_μ from LSND), and integrating differential cross sections for ν_μ in Eq. 24 and 27 from $T_{e,min}$ to $T_{e,max} \rightarrow 2 \text{ GeV}$ results in

$$\frac{\sigma_{\text{MAG}}}{\sigma_{\text{SM}}} \approx \begin{cases} 0.035 & T_{e,min} = 0.5 \text{ GeV}, \\ 0.14 & T_{e,min} = 0.01 \text{ GeV}. \end{cases} \quad (28)$$

Therefore, at the current NOvA detection capabilities, there are about 0.035 times as many neutrino magnetic moment ν -on-e events than SM ones. This can be compared with the expected statistical uncertainty on the SM background, which in the case of Poisson distributed events is the square root of the number of predicted events. Consequently, it is possible to assess the minimal number of SM ν -on-e events necessary for the magnetic moment signal to be detected above the SM background (without considering systematic uncertainties) as

$$N_{SM} > 1/0.035^2 \approx 816. \quad (29)$$

However, this approximation is calculated only for one value of E_ν , but can be used to assess the sensitivity of the experiment.

As can be seen in Fig. 4 and Fig. 5, the magnetic moment contribution exceeds the SM contribution for low enough T_e . This can be approximated as [15]:

$$T_e \lesssim \frac{\pi^2 \alpha^2}{G_F^2 m_e^3} \left(\frac{\mu_\nu}{\mu_B} \right)^2 \simeq 2.9 \times 10^{19} \left(\frac{\mu_\nu}{\mu_B} \right)^2 [\text{MeV}], \quad (30)$$

which does not depend on the neutrino energy. Therefore, experiments sensitive to lower energetic electrons are significantly more sensitive to the neutrino magnetic moment.

3 Analysis Overview

Our analysis strategy for measuring the effective muon neutrino magnetic moment in the NOvA ND is based on comparing the total number of reconstructed and selected events in data with the prediction. The predicted events consist of the signal, which depends on the size of the effective muon neutrino magnetic moment, and of the background, which corresponds to the SM-only (null) hypothesis without any neutrino magnetic moment. We define the signal as true ν -on-e elastic scattering interactions, created with the use of the neutrino magnetic moment cross section instead of the SM cross section, as described in Sec. 2.2. Additionally, the signal events are required to have their true interaction vertex contained within the ND to exclude events originating from outside of the detector.

The ν -on-e interactions are also used in the ND group's analysis to constraint the neutrino beam prediction [13], which relies on the precise theoretical knowledge of the ν -on-e interaction cross section. They compare the total number of recorded ν -on-e interactions, with background subtracted based on a

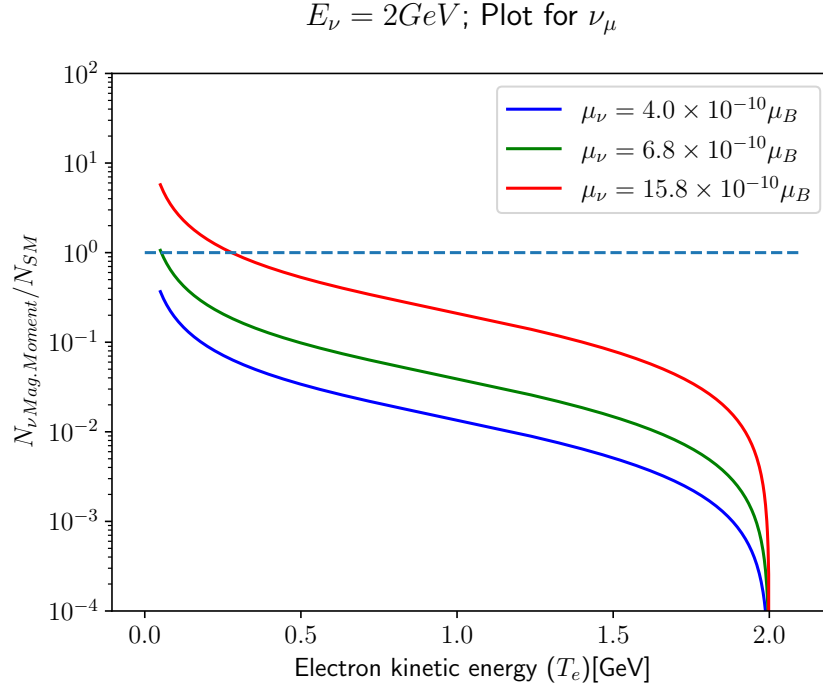


Figure 5: Ratio of the neutrino magnetic moment cross section to the SM cross section for the ν -on-e elastic scattering of 2 GeV ν_μ . Different colours depict different effective muon neutrino magnetic moment values, with red corresponding to the previous NOvA measurement, green the LSND result, and blue a possible ultimate NOvA sensitivity, as discussed in the introduction to this chapter.

data/Monte Carlo (MC) comparison in a sideband region, to the prediction. Since the number of ν -on-e events should only depend on the normalization of the neutrino beam, this analysis should give us a precise validation, or correction, of the neutrino flux normalization. There has been a large amount of work going into this analysis, including making special samples, weights, event classifiers, developing a dedicated event selection, or developing a background subtraction method, among others. To save time and analysis effort, we have taken most of these work at face value and applied it to the neutrino magnetic moment analysis. This will help us get the first good estimate of NOvA's capabilities to constraint (or measure) the neutrino magnetic moment.

The same detector signature of a single forward going electron shower, as present in the ν -on-e events, is also present in the LDM analysis [14]. This analysis is using a similar event selection to select the LDM events as the ND group, only without the final $E\theta^2$ cut (see Sec. 4). However, instead of simply comparing the total event counts, the LDM analysis is using a CAFAna-based fitting framework to fit for the possible LDM signal in a distribution of electron recoil energy multiplied by electron recoil angle squared.

The data used in this analysis were collected from the start of NOvA ND data taking on the August 22nd, 2014, until February 3rd, 2021. This is the ND data that were used in the latest NOvA neutrino oscillations result [25], with an additional year. Although more data have been collected since February 2021, they are still being processed and are not available at the time of writing this thesis. The total exposure of the data sample is approximately 13.8×10^{20} Protons On Target (POT). This exposure is used throughout this chapter to scale the predicted distributions and number of events.

For this analysis we are using near detector CAF samples with the standard Production 5.1 reconstruction. The SAMWEB definition for the data sample is:

prod_caf_R20-11-25-prod5.1reco.l_nd_numi_fhc_full_v1_goodruns.

We are following the standard data blinding procedure and have not looked at any data events until the analysis passes the full collaboration review.

We are using the standard NOvA Prod5.1 ND simulation, with Geant4 v4.10.4.p02 [26] used for the neutrino flux prediction and GENIE v3.0.6 [27] with tune N1810j_02_11a for the neutrino interactions. This is the GENIE version that experienced the ‘*Genie Skew*’ (TO DO: *Describe the genie skew fix properly and reference, maybe docdb:53691 or something newer*). The list of CAFAna simulation samples is in Tab. 1.

The simulation was created with approximately four times larger statistics than the data to limit statistical uncertainties from simulation with total exposure for the simulation being approximately 55.4×10^{20} POT. For the systematic uncertainty studies only a portion of this full sample is used, specifically 19.3×10^{20} POT.

Table 1: SAMWEB definitions of the nominal Near Detector Prod5.1 simulation samples.

Nominal:

prod_caf_R20-11-25-prod5.1reco.a_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_v1

Systematically shifted samples:

prod_caf_R20-11-25-prod5.1reco.e_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_calibup_v1_batch2

prod_caf_R20-11-25-prod5.1reco.e_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_calibdown_v1_batch2

prod_caf_R20-11-25-prod5.1reco.f_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_ckvup_v1_batch2

prod_caf_R20-11-25-prod5.1reco.f_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_ckvdown_v1_batch2

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_lightlevelup_v1_batch2

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_lightlevelup_v1_batch2

prod_caf_R20-11-25-prod5.1reco.f_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_detectorageing_v1_batch2

prod_caf_R20-11-25-prod5.1reco.f_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_calibshape_v1_batch2

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_mcnp_v1_batch2

3.1 Analysis Weights

Corrections for known limitations in the simulation are applied in the form of analysis weights applied to each event based on how it is affected by specific variations in the simulation. This includes the corrections for the neutrino beam prediction based on the external measurements used by the Package to Predict the Flux (PPFX), and, for the non- ν -on- e background only, also the internal and external measurements that constrain the neutrino interaction prediction inside GENIE.

Radiative Correction Weight

The cross section corrections are not applied to the ν -on- e events, as they are assumed to be known precisely from theory. However, the GENIE MC simulation only considers tree-level SM ν -on- e interactions [13], as described in Sec. 2.2, and doesn't account for any higher order terms, which are described by radiative corrections.

Radiative corrections can be expressed by two adjustments to the tree-level SM ν -on- e cross section [28]. First, the values of the weak coupling constants are changed as [29]

$$g_1^{\nu e} \rightarrow 0.7276, \quad g_1^{\nu \mu} \rightarrow -0.2730, \quad g_2 \rightarrow 0.2334. \quad (31)$$

Second, there are additional terms added to the cross section equation. Considering only one-loop corrections, the full ν -on- e cross section can be expressed as¹ [30]

$$\left(\frac{d\sigma_{\nu\text{-on-}e}}{dy} \right)_{\text{Rad. Corr.}} = \frac{G_F^2 s}{\pi} \left\{ g_1^2 \left(1 + \frac{\alpha}{\pi} X_1 \right) + g_2^2 (1-y)^2 \left(1 + \frac{\alpha}{\pi} X_2 \right) - g_1 g_2 \frac{m_e y}{E_\nu} \right\}, \quad (32)$$

where

$$y = \frac{T_e + E_\gamma}{E_\nu}, \quad (33)$$

$s = 2E_\nu m_e + m_e^2$ is the Mandelstam variable,

$$X_1 = -\frac{2}{3} \log \left(\frac{2yE_\nu}{m_e} \right) + \frac{y^2}{24} - \frac{5y}{12} - \frac{\pi^2}{6} + \frac{23}{72} \quad (34)$$

and

$$X_2 = -\frac{2}{3} (1-y)^2 \log \left(\frac{2yE_\nu}{m_e} \right) - \frac{y^2}{18} - \frac{\pi^2}{6} (1-y)^2 - \frac{2y}{9} + \frac{23}{72}. \quad (35)$$

In practice, we are using the implementation of radiative corrections as a weight, as created by Yiwen Xiao [13], where each true ν -on- e event is weighted by a ratio

$$\text{weight}_{\text{Rad. Corr.}}(E_\nu, T_e) = \left(\frac{d\sigma_{\nu\text{-on-}e}}{dy} \right)_{\text{Rad. Corr.}} / \left(\frac{d\sigma_{\nu\text{-on-}e}}{dy} \right)_{\text{GENIE 3}}; \quad (36)$$

Neutrino Magnetic Moment Signal as a Weight

Analogically to the radiative correction weight, it is possible to create a neutrino magnetic moment weight as a ratio between the neutrino magnetic moment and the SM differential cross sections for the ν -on- e interactions. This can then serve to predict the number of ν -on- e events created by the neutrino magnetic

¹There is technically a third correction term X_3 by the $g_1 g_2$ term, which is however negligible for $E_\nu \sim \text{GeV}$.

moment interaction (which make up the signal), without the need for an additional simulation. This is possible thanks to the theoretically very well understood properties of the ν -on-e interaction, as described in Sec. 2.2. Therefore, the signal sample is created from the true ν -on-e sample, with the magnetic moment weight applied. The weight has a form:

$$\text{weight}_{\nu\text{Mag. Mom.}}(E_\nu, T_e) = \left(\frac{d\sigma_{\nu\text{-on-e}}}{dy} \right)_{\nu\text{Mag. Mom.}} / \left(\frac{d\sigma_{\nu\text{-on-e}}}{dy} \right)_{\text{GENIE 3}}, \quad (37)$$

where

$$\left(\frac{d\sigma_{\nu\text{-on-e}}}{dy} \right)_{\nu\text{Mag. Mom.}} = E_\nu \left(\frac{d\sigma_{\nu\text{-on-e}}}{dT_e} \right)_{\nu\text{Mag. Mom.}}. \quad (38)$$

3.2 Special Simulation Samples

Enhanced ν -on-e Sample

Due to the relatively low cross section of the ν -on-e interaction, the nominal simulation sample contains very few ν -on-e events, which could result in a significant statistical uncertainty from simulation. To avoid this, we are using the ν -on-e-enhanced simulation sample created by Wenjie [13], which is mainly made up of ν -on-e events with a total exposure of 1.72×10^{24} POT. There are a few non- ν -on-e background events overlaid on top of the ν -on-e events to properly account for the possible reconstruction effects of the pileup of neutrino interactions in a single spill [13], since in the real detector, the hits from the true ν -on-e interaction can be clustered together into another interaction, or additional hits can be clustered together into the ν -on-e event. To save up on unnecessary disk space and processing usage, the enhanced ν -on-e sample does not include any cross section related parameters and variables, as the ν -on-e interaction is assumed to be known exactly from theory. Therefore, we do not apply cross section weights or account for cross section systematic uncertainties for ν -on-e events.

(TO DO: Describe the technical details of the *nuone* sample) We are using the overlaid and filematched samples for consistency. The CAF files have incorrect POT, which is $3.6995434\text{e}+20$. This needs to be corrected to their correct POT of $1.7209423\text{e}+24$ (this is for filematched sample). We only have systematic samples for the absolute calibration, cherenkov and light level. This is because the effect of ageing, calibration shape, and neutrons should be negligible for ν -on-e events.

The list of the *nu-on-e* sample definitions is in table 2.

Enhanced ν_e CC MEC Sample

The cross section tuning procedure in NOvA applies large weights to Meson Exchange Current (MEC) events in some parts of the parameter space. However, after the full event selection (Sec. 4) only a small number of MEC events remain in the detector. This was shown to be an issue especially for the ν_e CC MEC events [13]. Applying large tuning corrections to a small number of events results in large statistical fluctuations. To avoid this, we are using another special sample created by Yiwen Xiao [13] with an enhanced number of ν_e CC MEC events, following the same procedure as for the ν -on-e-enhanced sample, with an exposure of 1.99×10^{24} POT.

(TO DO: Describe the technical details) Incorrect POT is $4.7334120\text{e}+23$ and correct POT is $1.9880340\text{e}+24$. This is for filematched samples. There is a limitation of the sample in the $q3$ - $q0$ parameter space. Need to list it or just reference the technote.

The list of the ν_e CCMEC enhanced sample definitions is in table 3.

Table 2: SAMWEB definitions for the enhanced v -on- e samples.**Nominal:**

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_v1_nuone_overlay

Systematically shifted samples:

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_calibup_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_calibdown_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_ckvup_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_ckvdown_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_lightlevelup_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_lightleveldown_v1_nuone_overlay

Table 3: SAMWEB definitions of the v_e CC MEC background sample.**Nominal:**

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_filematch

Systematically shifted samples:

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CalibUp_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CalibDown_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CkvUp_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CkvDown_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_LLUUp_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_LLDnDown_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_Aging_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CalibShape_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_MCNP_Combined_240124_filematch

272 A summary of the simulation samples and analysis weights for the four different types of signal and
273 background components is shown in Tab. 4. In the following chapter, the v_e CC MEC background is
274 added into the ‘Other background’ sample, even though it is created from a separate simulation.

Table 4: Overview of the simulation samples and analysis weight used for the different signal and background components.

Signal type	Sample	Weight
Signal	Enhanced ν -on-e	Flux & ν Mag. Moment
ν -on-e background	Enhanced ν -on-e	Flux & Rad. Corr.
ν_e CC MEC background	Enhanced ν_e CC MEC	Flux & Cross Sec.
Other background	Nominal ND	Flux & Cross Sec.

4 Event Selection

We are searching for ν -on-e elastic scattering events, characterised by a single very forward going electron shower, specifically focusing on low electron recoil energies. The main backgrounds for our analysis come from ν_e CC interactions, which produce an electron with additional activity, and interactions that produce π^0 , which decays into two photons producing electromagnetic showers, where each can look similar to the ν -on-e signal. Additionally, there are ν_μ CC interactions, which are generally easy to distinguish from our signal, however, their very high abundance in the NOvA ND makes them a dominant background nevertheless.

The signal definitions use the `kMode` variable instead of the `kIntType`, which is now deprecated (ref.: various Slack conversations). Mode 10005 denotes only the ν -on-e events and is only available for the enhanced ν -on-e samples, while mode 5 denotes all electron scattering events, including inverse muon decay interactions. That is why we had to add a requirement of an electron in the final state. Mode 10 denotes all Meson Exchange Current (MEC) events. We are using the ND group's [13] signal definition including a requirement for the true vertex to lie within their fiducial volume (defined below). This is the cut name `NDNuoneFiducial`. This is in contrast with the LDM analysis [14], which uses the `ana::kVtxIsContained` cut instead, which has a looser boundary. This choice has only a negligible effect on the final number of selected events and only affects the selection efficiency.

We explain the motivation behind each cut of the event selection and discuss their effect on the neutrino magnetic moment events below. We also consider possible improvements to the event selection for a future (re-)analysis.

The strategy for event selection is as follows. First, we remove events that failed reconstruction or data collection, described in Sec. 4.1 and 4.2. Then, we apply pre-selection cuts that remove obvious background (Sec. 4.3), while limiting the reduction of the signal efficiency to about 0.25 %. Following this, we apply the containment cuts (Sec. 4.4) that remove events that are either not fully contained within the detector, or events that originate from outside of the detector, such as rock muons. Afterwards, we perform a cut-based Multi Variate Analysis (MVA) on a selection of variables useful for distinguishing the signal from the background, discussed in Sec. 4.5, and evaluate their combined performance on the signal selection. We choose the cut values that result in the best statistical significance, based on a chosen Figure Of Merit (FOM). Given that we are searching for a very limited number of signal events on top of a large background, we chose a simple statistics-only FOM

$$\text{FOM} = \frac{\text{Signal}}{\sqrt{\text{Background}}}. \quad (39)$$

The summary of the cut values for the event selection of neutrino magnetic moment signal is presented

in Tab. 5, showing the label for the event selection variable, its description and the cut value chosen. After the full event selection, the predicted number of signal events for $\mu_\nu = 10^{-9}\mu_B$ is 56.80 and the total number of background events under the SM hypothesis is 700.33.

Table 5: Summary of the variables and their cut values for the event selection of neutrino magnetic moment signal. Showing the category of the event selection variable, its label, description and the cut value chosen.

	Label	Description	Cut
Reco Qual.	Valid Vtx	Valid reconstructed vertex	> 0
	N^o Prongs	Number of reconstructed prongs	> 0
	Hits / Plane	Number of hits per plane	< 6
	Low E_{Shower}	Low cut on calorimetric energy of the most energetic shower	$> 0.5 \text{ GeV}$
Preselection	N^o Hits Loose	Preliminary cut on the total number of hits for all prongs in a slice	< 280
	Prong Length	Length of the longest prong	$< 640 \text{ cm}$
	$E\theta^2$ Loose	Preliminary cut on the product of the calorimetric energy and angle squared of the leading shower	$< 0.064 \text{ GeV} \times \text{rad}^2$
Fiducial	Vertex	x position	$> -177 \text{ cm}$ $< 177 \text{ cm}$
		y position	$> -177 \text{ cm}$ $< 177 \text{ cm}$
		z position	$> 50 \text{ cm}$ $< 1170 \text{ cm}$
Containment	Prong	Minimum hit position in x	$> -177 \text{ cm}$
		Maximum hit position in x	$< 177 \text{ cm}$
		Minimum hit position in y	$> -185 \text{ cm}$
		Maximum hit position in y	$< 177 \text{ cm}$
		Minimum hit position in z	$> 55 \text{ cm}$
		Maximum hit position in z	$< 1270 \text{ cm}$
Selection	E_{Shower}/E_{Tot}	Fraction of energy contained in the most energetic shower	> 0.91
	N^o Hits	Total number of hits for all prongs in a slice	< 116
	High E_{Shower}	Calorimetric energy of the most energetic shower	$< 1.4 \text{ GeV}$
	ν-on-e ID	Convolutional Visual Network (CVN)-based ν -on-e identifier	> 0.65
	$E\pi^0$ ID	CVN-based ν -on-e and π^0 identifier	> 0.63
	$E\theta^2$	Product of the calorimetric energy and angle squared of the leading shower	$< 0.0048 \text{ GeV} \times \text{rad}^2$

The code to make plots and analyse the event selection is located in the NuMagMomentAna/NuMMAAnalysis/EventSelection directory. The signal definition is in NuoneCuts.h and the cut values in the EventCVNCuts.h/cxx scripts from the NuMagMomentAna/Cuts directory.

4.1 Data Collection Quality

To ensure good data quality, we apply the following criteria to data (not applied to simulation) [31]. A cut on the time of each spill relative to other spills and on the exposure of each spill, where every spill is required to have at least 2^{12} POT. Additionally, the current in the focusing horn is required to be within $-202\text{ kA} < I_{Horn} < -196.4\text{ kA}$, the position of the beam to be within $\pm 2\text{ cm}$ in both x and y axis, and that the width of the beam to be within 0.57 and 1.58 cm. Furthermore, incomplete events, or events with issues in one or more Data Concentration Modules (DCMs) are removed.

4.2 Reconstruction Quality

Electrons are reconstructed by slicing, followed by vertexing, then clustering into prongs. To identify electrons we require a valid reconstructed vertex and at least one reconstructed prong. Even though electrons only consist of a single shower, we don't reject events with more than one prong in a slice, as the reconstruction can wrongly assign noise hits as a separate prong. These false secondary prongs can be removed later in the event selection.

Figure 6 and Tab. 6 show that about 68 % of signal events do not have a valid reconstructed vertex. This is due to the concentration of signal events at very low electron recoil energies, which results in events that can consist of a small number of hits, or even a single hit. As can be seen in the bottom plot in Fig. 6, events with small true electron recoil energies have much smaller vertex reconstruction efficiency than the higher energetic electrons. However, ongoing work is improving the vertex reconstruction in the NOvA detectors with a use of Machine Learning (ML) instead of the currently used Hough transform combined with Elastic Arms [32]. Improving NOvA vertex reconstruction at low energies can enhance our event selection in the future.

Additionally, we limit the number of hits per plane to < 6 . This is to remove the so-called 'Front End Board (FEB) flashers', which are caused by such a high energy deposit in one cell, that it affects all the other channels on the same Avalanche Photodiode (APD) [33]. The cut value was chosen so that it removes approximately 0.25 % signal events, which is the same criterion as is used for the pre-selection cuts described below. Relative comparisons between signal and background for the number of prongs and the number of hits per plane are shown in Fig. 7.

Furthermore, the reconstructed calorimetric energy of the primary shower is required to be $E_{cal} > 0.5\text{ GeV}$ as shown in Fig. 8. This is primarily due to the limitations of the currently used CVN-based ν -on-e identifiers described in Sec. 4.5, which were developed and validated for ν -on-e events with energies above this limit, to avoid the large background at low energies. However, due to the nature of the neutrino magnetic moment signal, which is concentrated at low electron recoil energies, this cut also removes a majority of our signal events, specifically 66.8 %. This large reduction severely impacts the significance of our measurement. On the other hand, it also marks potentially the most impactful improvement available in a future re-analysis. There are other event identifying algorithms available in NOvA that could be explored for ν -on-e events to leverage the low energy sample. Additionally, it is possible to develop a purpose-built ν -on-e identifier focusing on low electron recoil energies.

4.3 Pre-Selection

Pre-selection aims to remove obvious background events without significantly affecting the signal. The criterion we chose for the selection of these cuts is determined by the reduction of the signal efficiency by approximately 0.25 % with each cut. This results in the total pre-selection reduction of the signal efficiency by approximately 1 %.

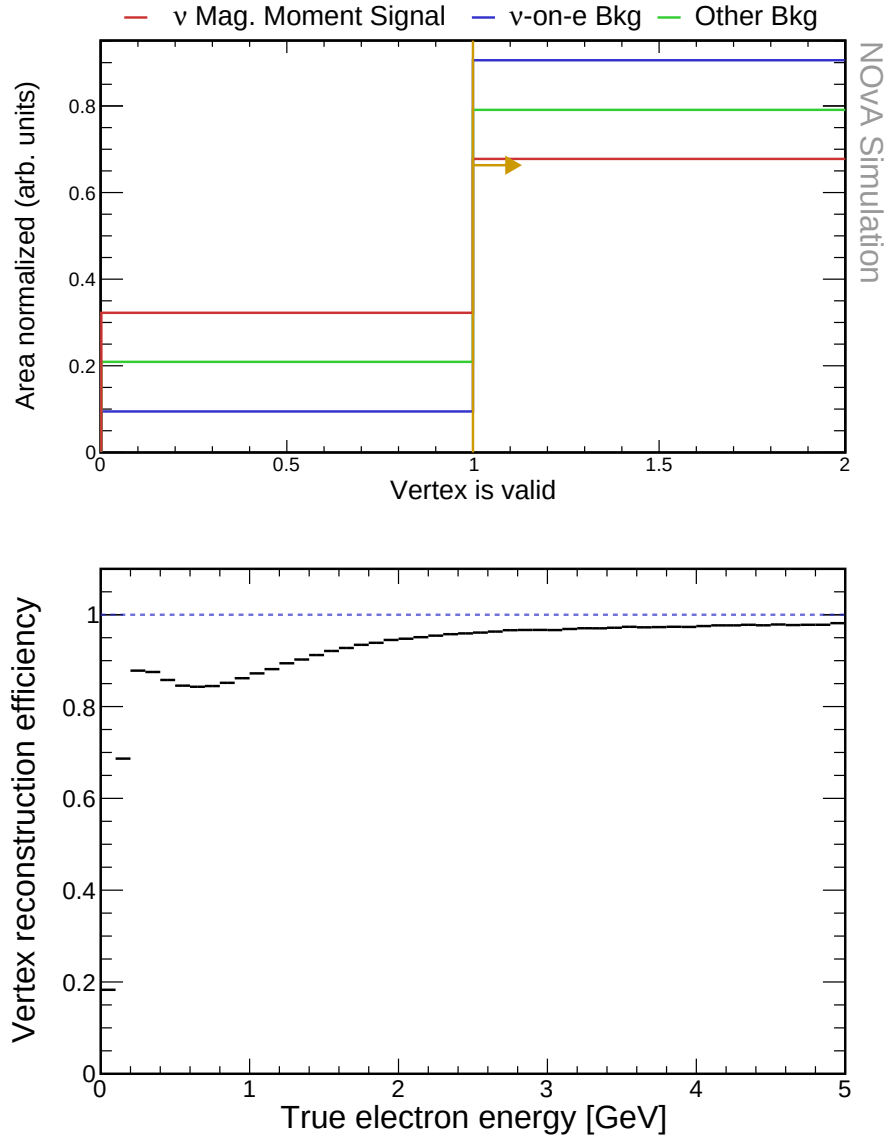


Figure 6: Top: Relative comparison of the signal (red), ν -on-e background (blue), and other background (green) events for the vertex reconstruction quality selection. Each histogram is area-normalised and the first bin corresponds to events without a valid vertex and second bin to events with correctly reconstructed vertex. The yellow line indicates the chosen cut value, where all events have to have a valid reconstructed vertex. Bottom: profile histogram of the 'vertex is valid' variable as a function of the true electron energy for the true signal events, showing the significant drop in vertex reconstruction efficiency at low electron recoil energies. No selection was applied prior to making these plots.

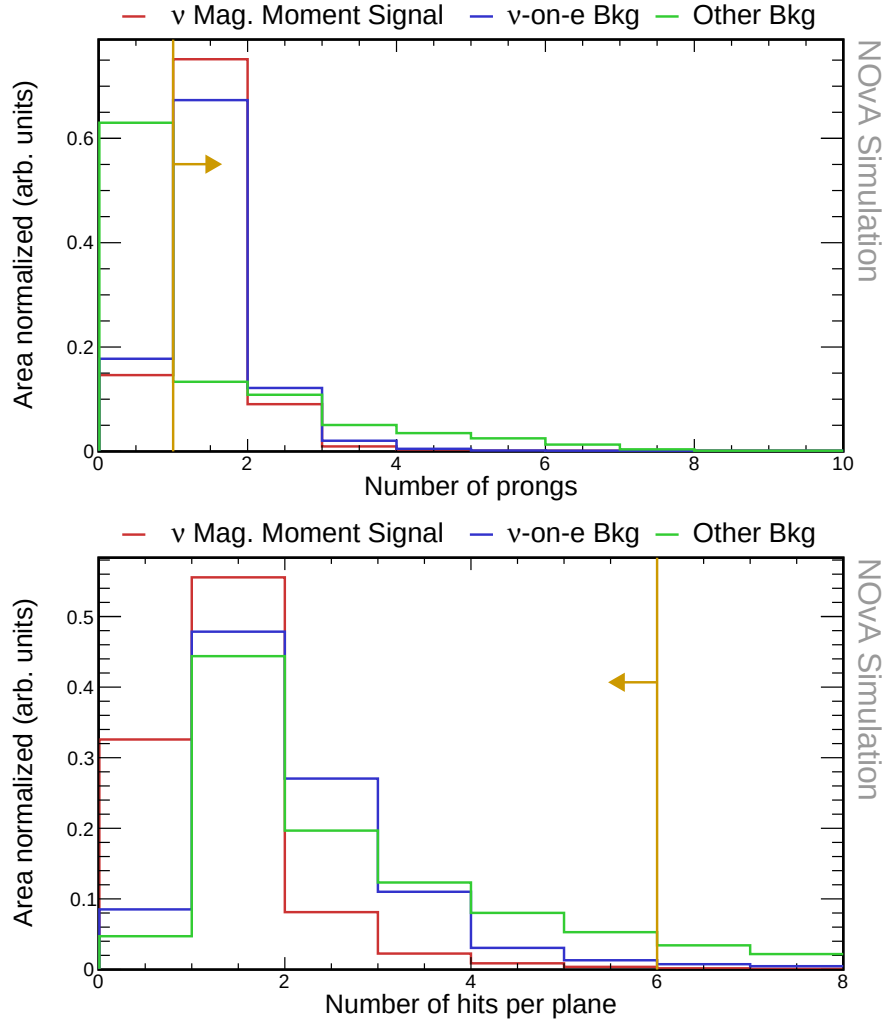


Figure 7: Relative comparison of signal (red), ν -on- e background (blue), and other background (green) events in the number of prongs (top) and the number of hits per plane (bottom) distributions. Events in both plots are required to have a valid reconstructed vertex and in the bottom plot also at least one reconstructed prong. Yellow lines indicate the cut values for the shown variables, with arrows pointing towards the preserved events. All histograms are area-normalised.

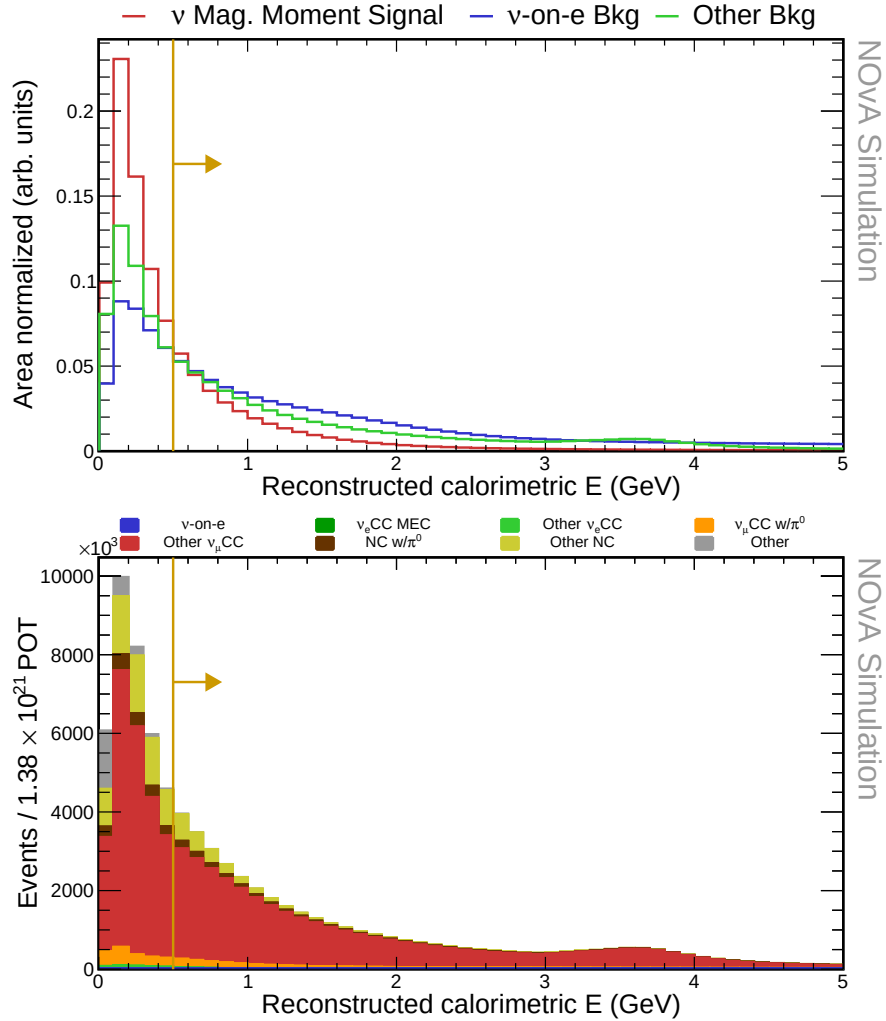


Figure 8: Top: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the reconstructed calorimetric energy distribution. All histograms are area-normalised. Bottom: Decomposition of background into various sub-samples, normalised to the data POT exposure. Events in both plots are required to have a valid reconstructed vertex, at least one reconstructed prong and less than 6 hits per plane. Yellow lines indicate the cut value for the reconstructed calorimetric energy, with arrows pointing towards the preserved events.

The first two variables used for our pre-selection are the same as were used in the event selection for the ν_e appearance ND constraint for the three flavour neutrino oscillation measurements [25]. As we are searching for single electron showers, we can reduce backgrounds with multiple final state particles by limiting the total activity in the detector. Specifically, we require that the total number of hits assigned to all the reconstructed prongs is < 280 . This is shown in Fig. 9. In general, the main background in NOvA consists of the ν_μ CC interactions, which are characterised by long muon tracks. We therefore limit the length of the longest reconstructed prong to be < 640 cm, as shown in Fig. 10.

Additionally, as discussed in Sec. 2.2, simple $2 \rightarrow 2$ kinematics dictate that the true electron recoil energy and angle for the ν -on-e interaction are limited by $E\theta^2 < 2m_e$. This can be used to distinguish ν -on-e elastic scattering from ν_e CC interactions, which also have an electron in the final state. However, due to unavoidable reconstruction deficiencies, the reconstructed $E\theta^2$ does not have such a strict cut-off value, and we are placing the pre-selection cut at $E\theta^2 < 0.064$, as can be seen in Fig. 11. Furthermore, some of the signal events can be reconstructed with the opposite direction with respect to the beam, which would result in $\theta \approx \pi$ rad. However, this reconstruction failure likely does not impact other reconstructed qualities and these events should be preserved for the final sample. For that reason, we are calculating the angle between the outgoing electron and the neutrino beam direction as $\arccos(\text{abs}(\cos \theta))$, which gives the same value whether the shower is reconstructed forward or backwards.

The effect of the pre-selection cuts on the signal and background samples are summarised in Tab. 7, where the first row lists the number of events after applying all the reconstruction quality cuts from Sec. 4.2. All three of the variables used for the pre-selection are employed again in the MVA, as described in Sec. 4.5.

4.4 Fiducial and Containment Cuts

To ensure all the deposited energy of the recoil electron is contained within the detector and to remove events originating outside of the detector (such as rock muons for the ND), we constrain the position of the reconstructed vertex and all the prongs in the slice. The decision on where to place the exact cut values is made based on the maximum FOM value.

The reconstructed vertex is required to be within the fiducial volume, which represents a well-understood volume of the detector. To select the fiducial volume, we investigate distributions of the reconstructed vertex in the x, y and z direction, shown in Fig. 12, 13 and 14 respectively. Basic reconstruction quality and pre-selection cuts are applied to make these distributions. Additionally, for the x and y position distributions, we require that the vertex is not placed inside of the Muon Catcher by requiring

Table 6: Event selection cutflow table for the reconstruction quality cuts showing the number of events and the relative efficiency of each cut for each signal sample. The relative efficiency is calculated as number of events remaining after applying the corresponding cut divided by number of events for all the previous cuts. All the cuts are listed in sequence as they are applied.

Selection	Signal		ν -on-e bkg		Other bkg	
	N_{evt}	$\epsilon_{rel} (\%)$	N_{evt}	$\epsilon_{rel} (\%)$	N_{evt}	$\epsilon_{rel} (\%)$
No Cut	817.34	100	6.82×10^3	100	2.96×10^8	100
Valid Vtx	553.86	67.76	6.17×10^3	90.55	2.34×10^8	79.10
N^o Prongs	472.90	85.38	5.08×10^3	82.25	8.66×10^7	37.00
Hits / Plane	471.14	99.63	4.97×10^3	97.85	7.32×10^7	89.56
Low E_{Shower}	156.37	33.19	3.53×10^3	71.09	4.06×10^7	55.12

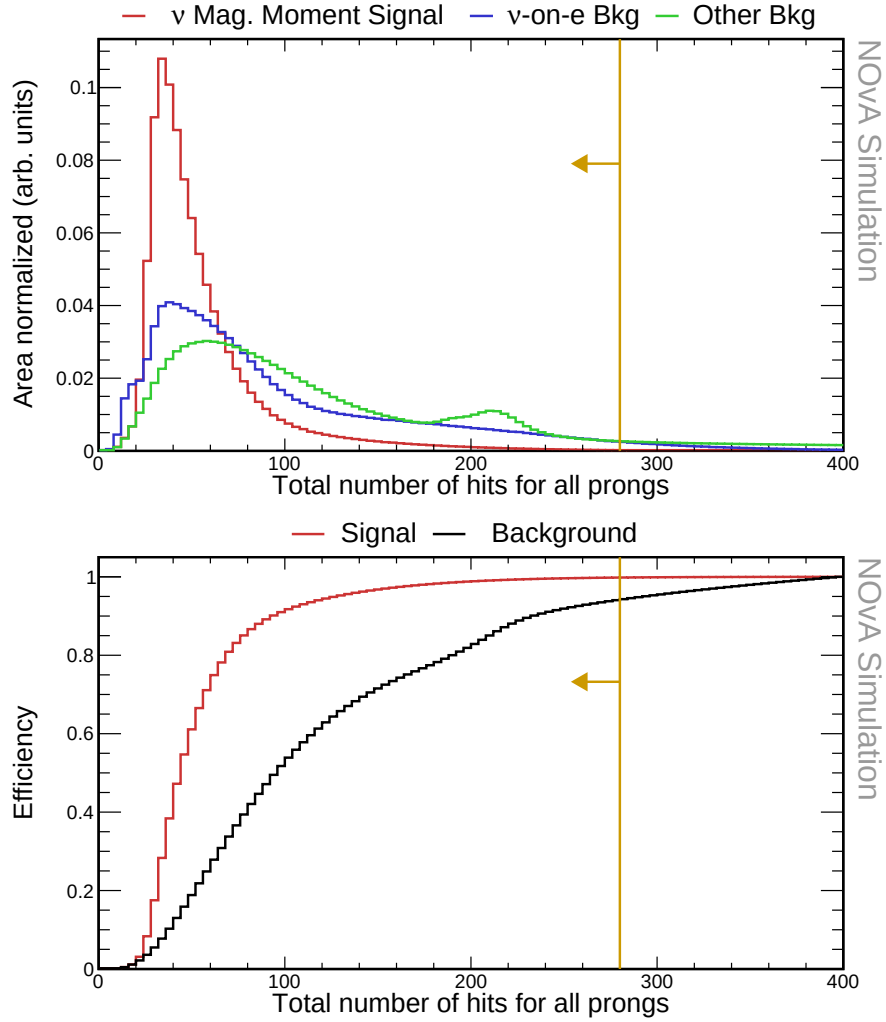


Figure 9: Top: Relative comparison of signal (red), ν -on- e background (blue), and other background (green) events in the distribution of total number of hits from all reconstructed prongs in the slice. All histograms are area-normalised. Bottom: Cumulative signal (red) and background (black) efficiency calculated as number of signal/background events left of the bin divided by the total number of signal/background events. Yellow lines indicate the cut value for the maximum number of hits, with arrows pointing towards the preserved events. The reconstruction quality cuts were applied before making these plots.

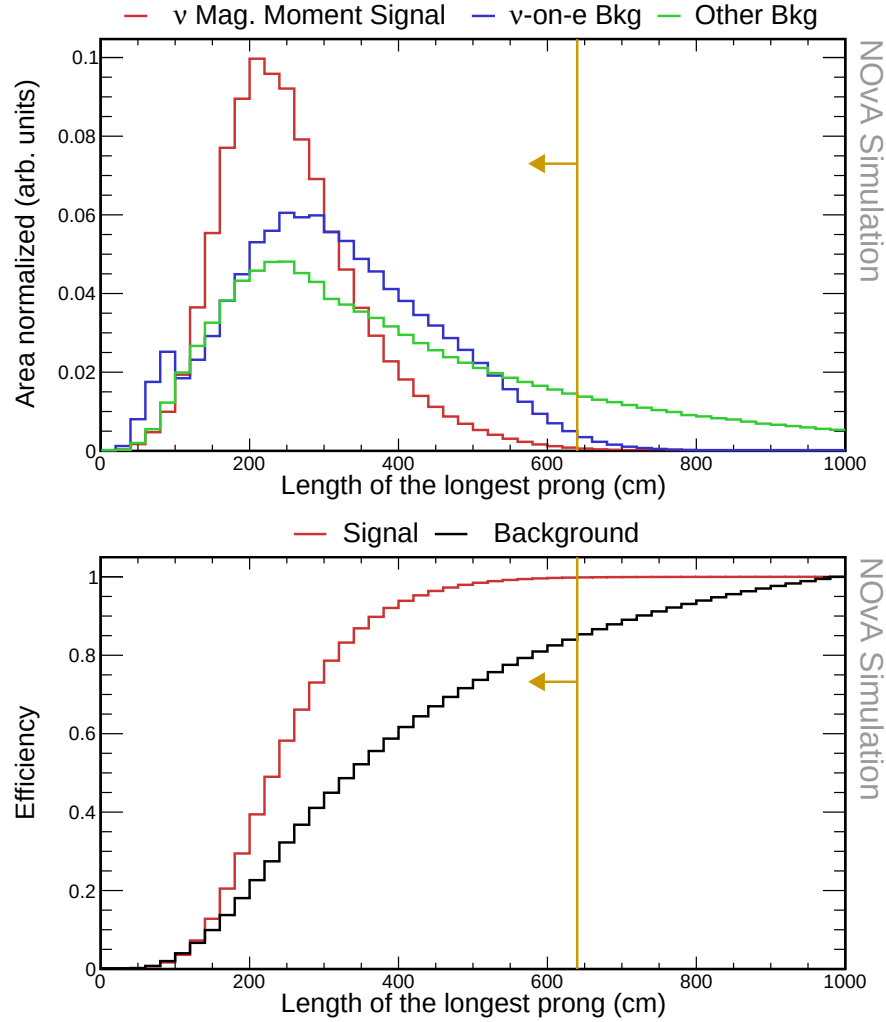


Figure 10: Top: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the length of the longest reconstructed prong in slice. All histograms are area-normalised. Bottom: Cumulative signal (red) and background (black) efficiency calculated as number of signal/background events left of the bin divided by the total number of signal/background events. Yellow lines indicate the cut value for the maximum length of the longest prong, with arrows pointing towards the preserved events. The reconstruction quality cuts and the number of hits cut were applied before making these plots.

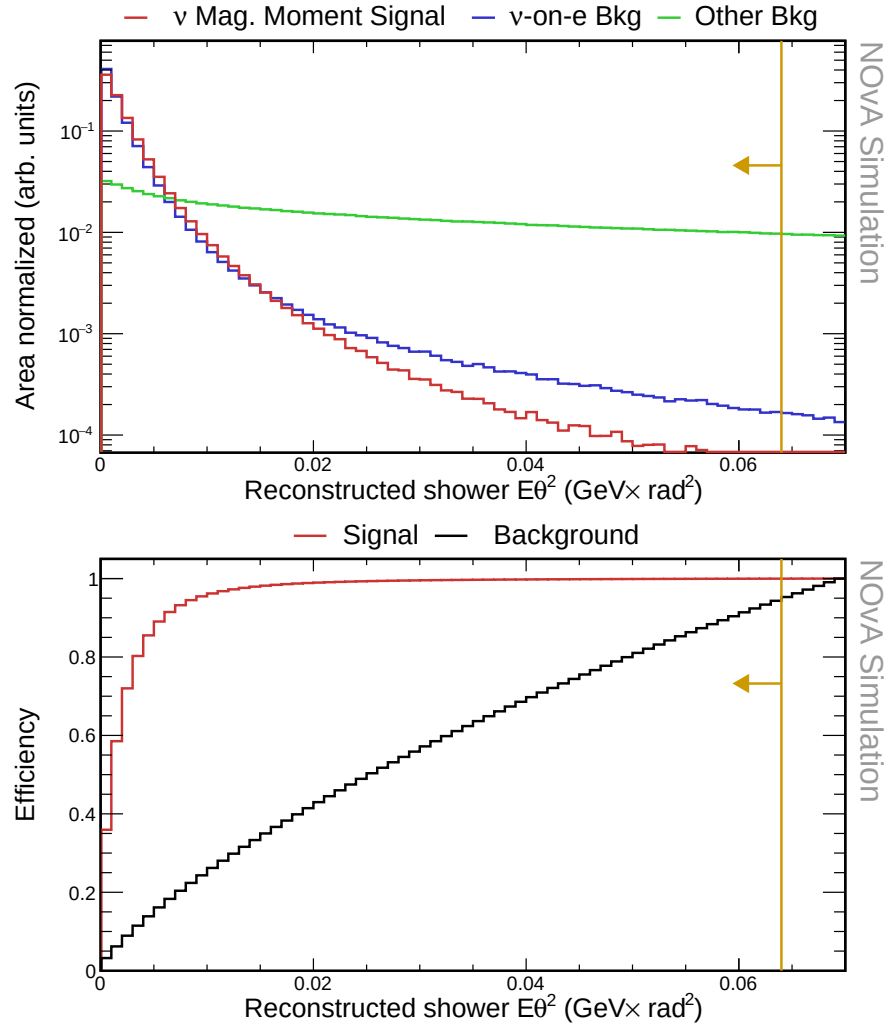


Figure 11: Top: Relative comparison of signal (red), ν -on- e background (blue), and other background (green) events in the distribution of the reconstructed energy of the leading shower multiplied by its angle from the incoming neutrino beam direction squared. All histograms are area-normalised with logarithmic y axis. Bottom: Cumulative signal (red) and background (black) efficiency calculated as number of signal/background events left of the bin divided by the total number of signal/background events. Yellow lines indicate the cut value for the depicted variable, with arrows pointing towards the preserved events. The reconstruction quality cuts, the number of hits cut, and the length of the longest prong cuts were applied before making both of these plots.

$V_{txZ} < 1270 \text{ cm}$, as it can significantly affect these distributions. The slanted distributions in x and y are caused by the off-axis nature of the Neutrinos from the Main Injector (NuMI) beam and the periodic peaks are due to a combination of the detector structure and the choice of binning.

The reconstructed vertex is required to be contained within the following volume:

$$-175 \text{ cm} < V_{txX} < 175 \text{ cm}, \quad (40)$$

$$-175 \text{ cm} < V_{txY} < 175 \text{ cm}, \quad (41)$$

$$95 \text{ cm} < V_{txZ} < 1170 \text{ cm}. \quad (42)$$

Furthermore, we constrain the extreme positions (minimum and maximum) of all the hits within the most energetic prong, which is assumed to represent the electron shower for the signal events. We apply the reconstruction quality, pre-selection and fiducial (vertex position) cuts to their distributions, shown in Fig. 15-20. The extreme hit positions are required to be within the following volume:

$$-175 \text{ cm} < \min_X, \max_X < 175 \text{ cm}, \quad (43)$$

$$-175 \text{ cm} < \min_Y, \max_Y < 175 \text{ cm}, \quad (44)$$

$$105 \text{ cm} < \min_Z, \max_Z < 1270 \text{ cm}. \quad (45)$$

4.5 Multivariate Analysis Cuts

Following the removal of obvious backgrounds and events not contained within the detector, we aim to optimise the event selection to achieve the highest significance for measuring the effective muon neutrino

Table 7: Pre-selection cutflow table showing the number of events and the relative efficiency of each cut for each signal sample. The relative efficiency is calculated as number of events remaining after applying the corresponding cut divided by number of events for all the previous cuts. All the cuts are listed in sequence as they are applied. The top row corresponds to the sample after applying the reconstruction quality cuts.

Selection	Signal		ν -on-e bkg		Other bkg	
	N_{evt}	$\epsilon_{rel} (\%)$	N_{evt}	$\epsilon_{rel} (\%)$	N_{evt}	$\epsilon_{rel} (\%)$
Reco Quality	156.37	100	3.53×10^3	100	4.28×10^7	100
N° Hits Loose	156.05	99.79	3.41×10^3	96.46	3.61×10^7	84.35
Prong Length	155.7	99.78	3.37×10^3	98.85	2.61×10^7	72.36
$E\theta^2$ Loose	155.14	99.64	3.33×10^3	98.83	8.83×10^6	33.82

Table 8: Event selection cutflow table for the containment cuts showing the number of events and the relative efficiency of each cut for each signal sample. The relative efficiency is calculated as number of events remaining after applying the corresponding cut divided by number of event for all the previous cuts. All the cuts are listed in sequence as they are applied. The top row corresponds to the sample after applying the reconstruction quality and pre-selection cuts.

Selection	Signal		ν -on-e bkg		Other bkg	
	N_{evt}	$\epsilon_{rel} (\%)$	N_{evt}	$\epsilon_{rel} (\%)$	N_{evt}	$\epsilon_{rel} (\%)$
Pre-selection	155.14	100	3.33×10^3	100	8.83×10^6	100
Fiducial	143.02	92.19	2.88×10^3	85.60	5.96×10^6	67.57
Containment	117.41	82.09	2.08×10^3	72.12	1.10×10^6	18.38

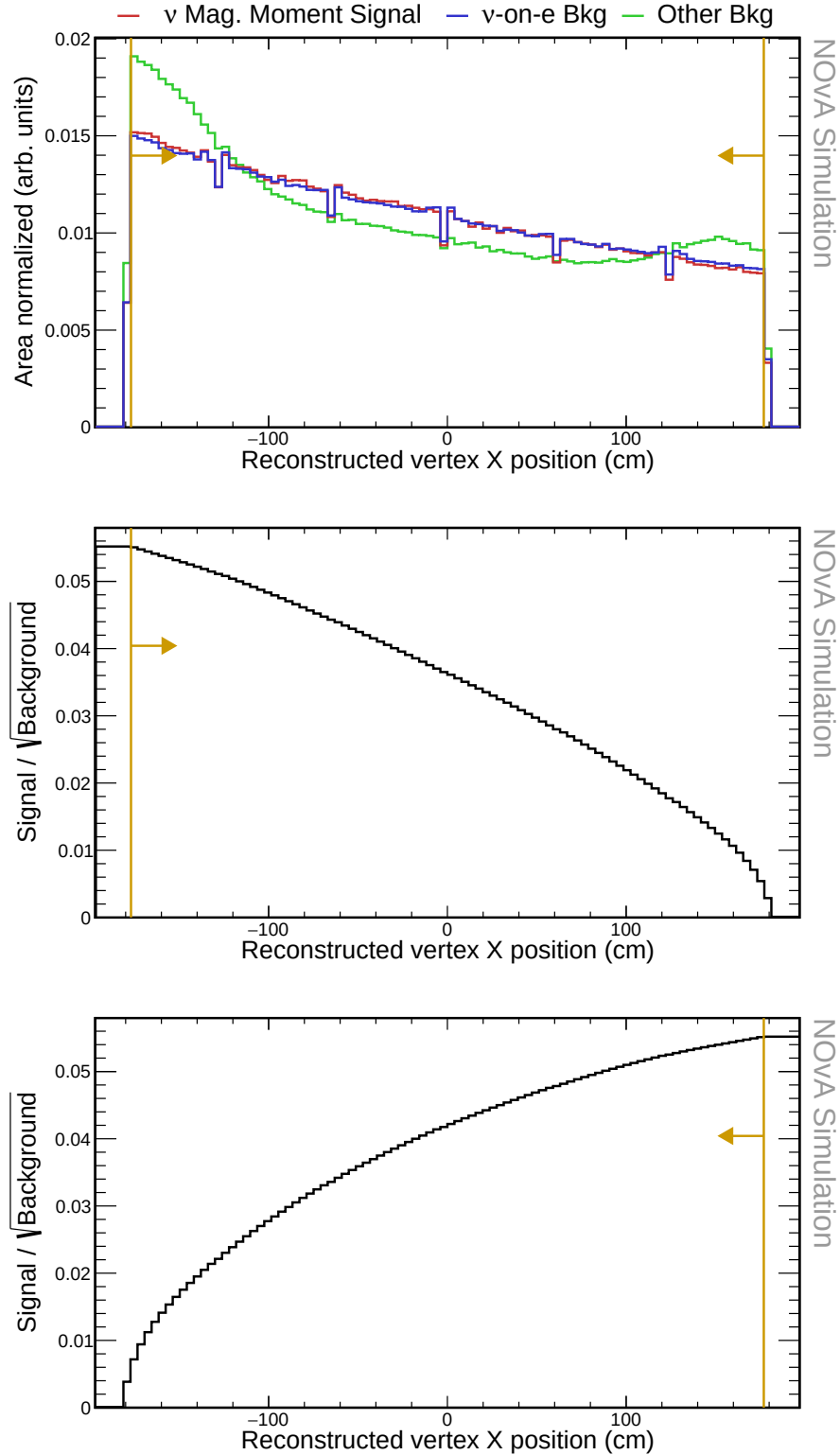


Figure 12: Top: Relative comparison of signal (red), ν -on- e background (blue), and other background (green) events in the distribution of the x position of the reconstructed vertex. All histograms are area-normalized. Middle and bottom: Cumulative FOM calculated as the number of signal events, divided by the number of background events from that bin until the end of the plot to the right (middle) or left (bottom). The reconstruction quality and pre-selection cuts were applied prior to making these plots. Additionally, vertex is required to be within the active region of the detector ($Vtx_Z < 1270\text{cm}$). Yellow lines show the cut values that create the fiducial volume, with arrows pointing towards the preserved events.

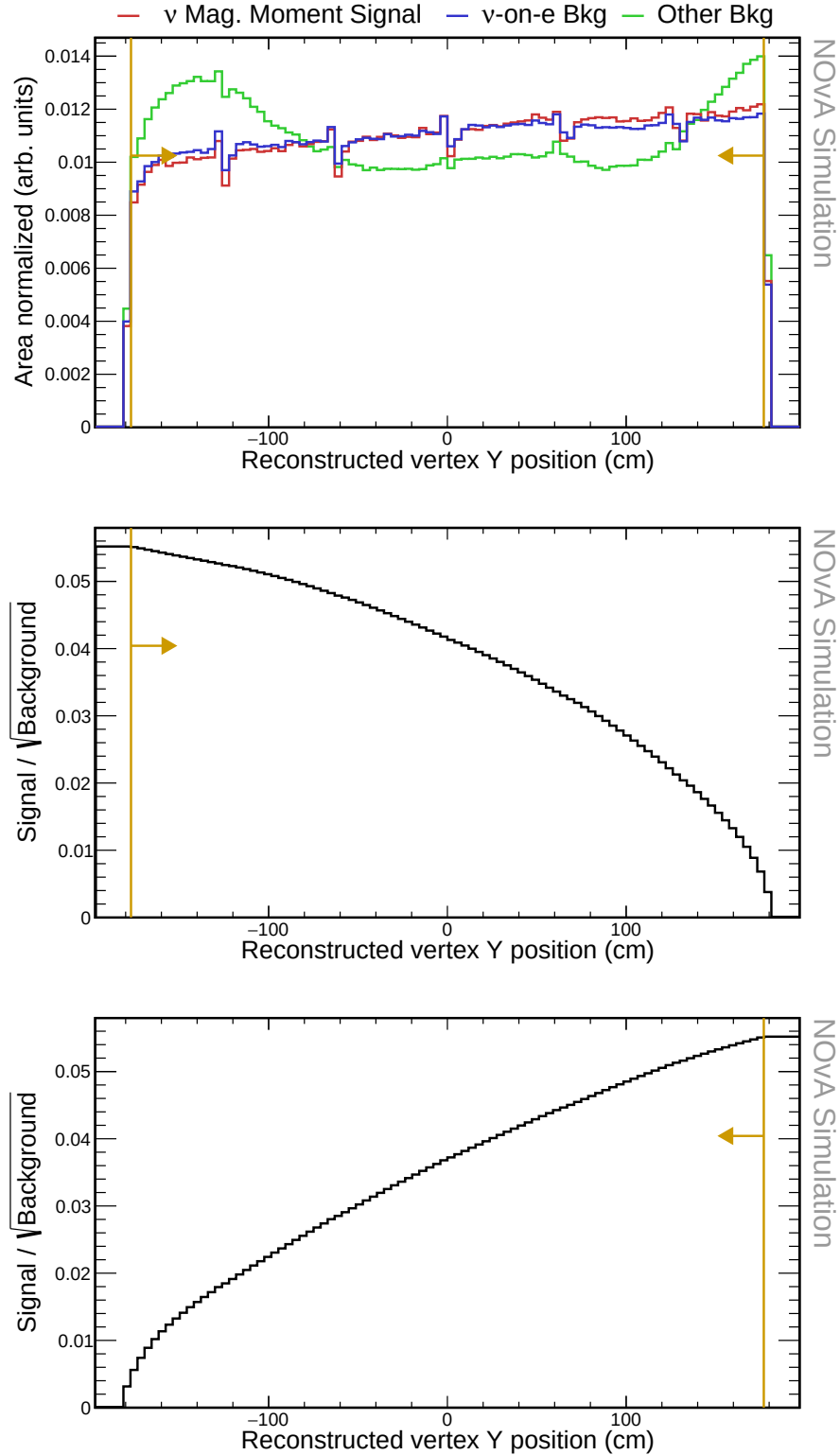


Figure 13: Top: Relative comparison of signal (red), ν -on- e background (blue), and other background (green) events in the distribution of the y position of the reconstructed vertex. All histograms are area-normalized. Middle and bottom: Cumulative FOM calculated as the number of signal events, divided by the number of background events from that bin until the end of the plot to the right (middle) or left (bottom). The reconstruction quality and pre-selection cuts were applied prior to making these plots. Additionally, vertex is required to be within the active region of the detector ($Vtx_Z < 1270\text{cm}$). Yellow lines show the cut values that create the fiducial volume, with arrows pointing towards the preserved events.

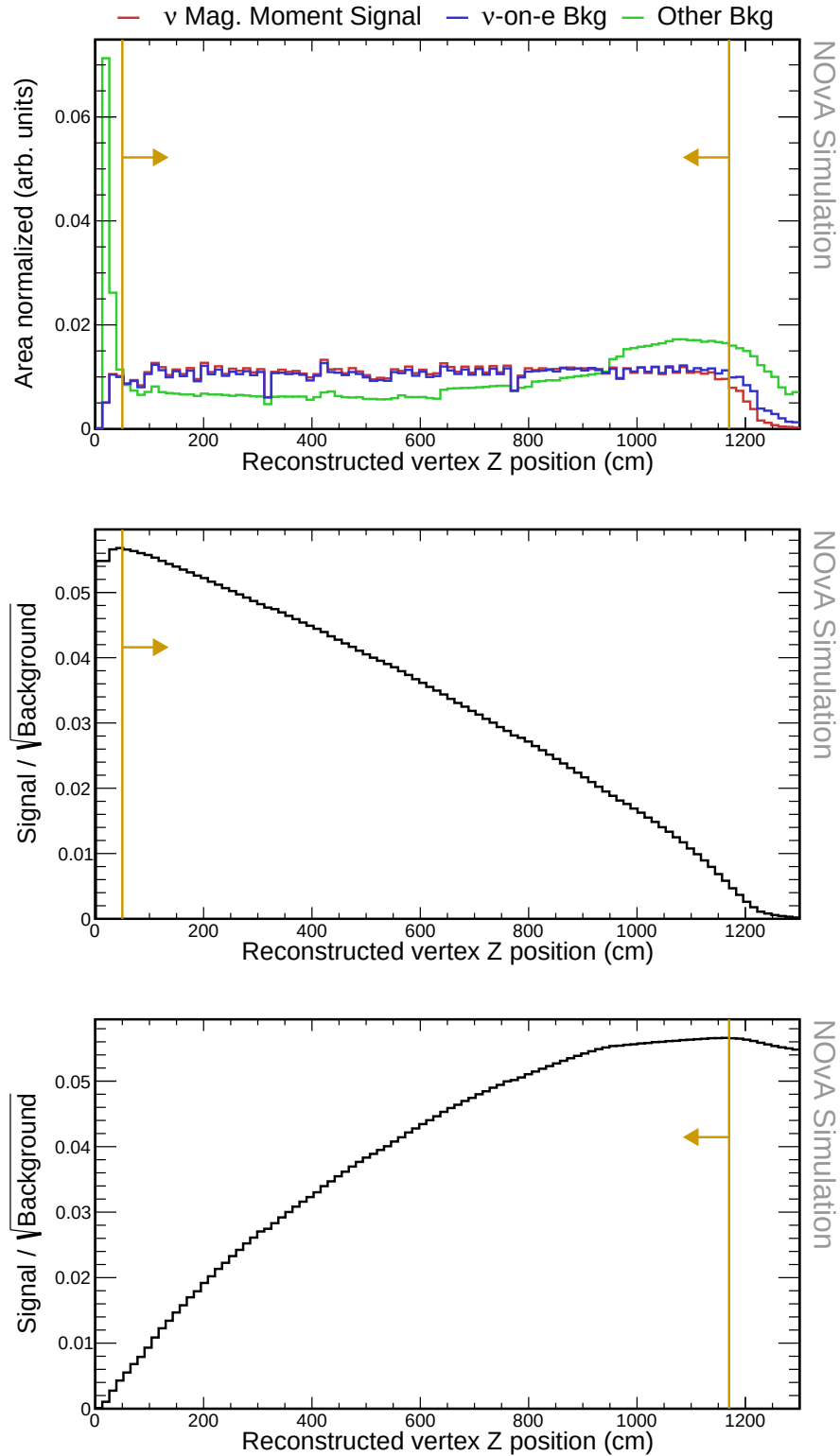


Figure 14: Top: Relative comparison of signal (red), ν -on- e background (blue), and other background (green) events in the distribution of the z position of the reconstructed vertex. All histograms are area-normalized. Middle and bottom: Cumulative FOM calculated as the number of signal events, divided by the number of background events from that bin until the end of the plot to the right (middle) or left (bottom). The reconstruction quality and pre-selection cuts were applied prior to making these plots. Yellow lines show the cut values that create the fiducial volume, with arrows pointing towards the preserved events.

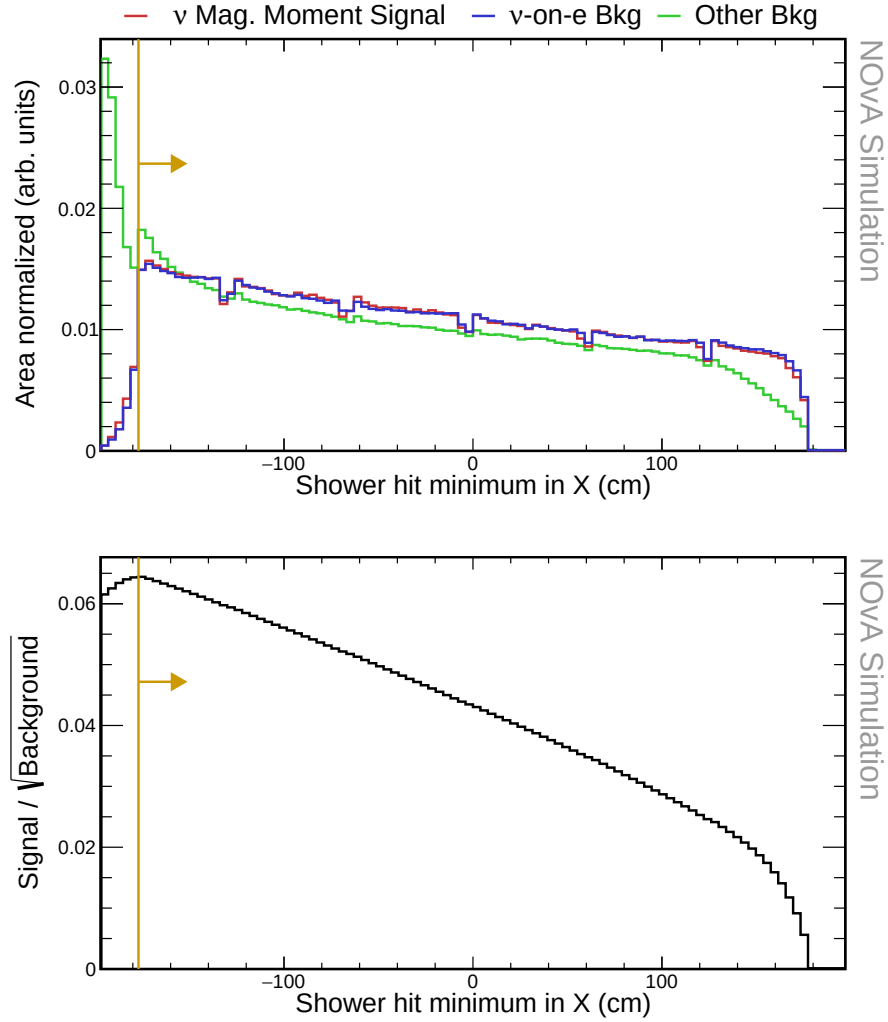


Figure 15: Top: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the minimum hit position of the most energetic prong along the x axis. All histograms are area-normalized. Bottom: Cumulative FOM calculated as the number of signal events, divided by the number of background events from that bin until the end of the plot in the direction of the yellow arrow. The reconstruction quality, pre-selection and fiducial cuts were applied prior to making these plots. Yellow lines show the cut values that create the containment volume, with arrows pointing towards the preserved events.

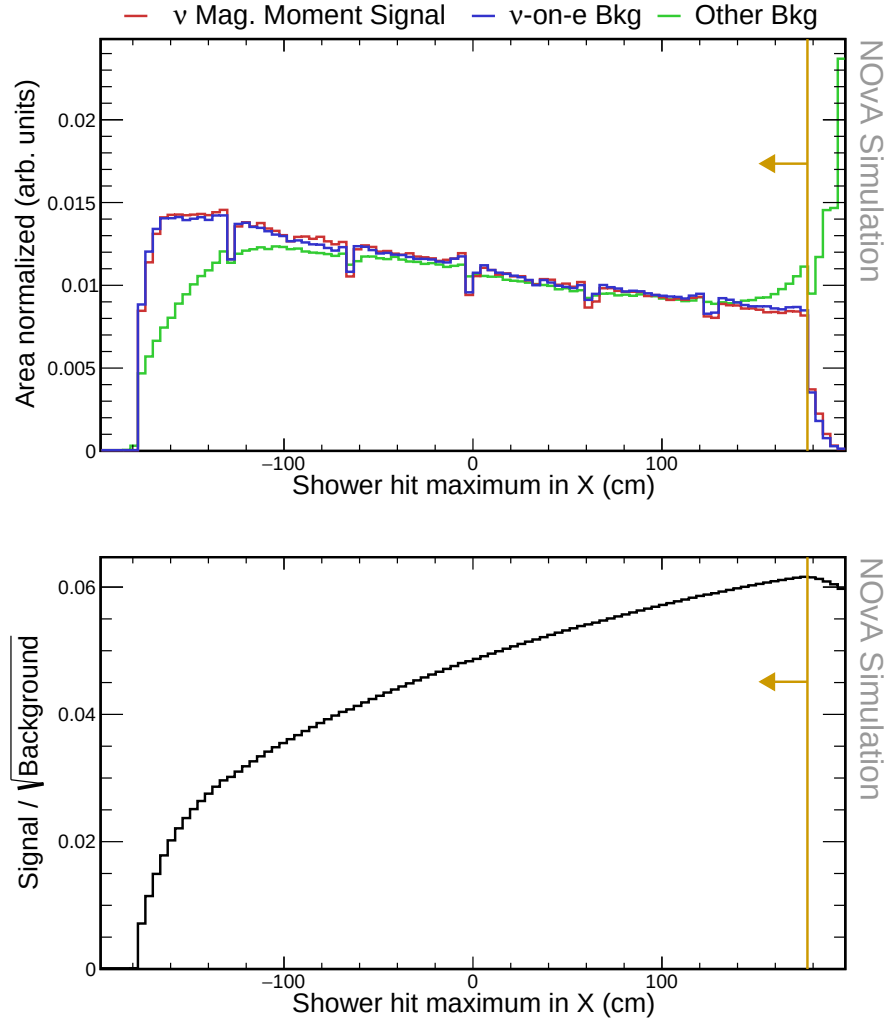


Figure 16: Top: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the maximum hit position of the most energetic prong along the x axis. All histograms are area-normalized. Bottom: Cumulative FOM calculated as the number of signal events, divided by the number of background events from that bin until the end of the plot in the direction of the yellow arrow. The reconstruction quality, pre-selection and fiducial cuts were applied prior to making these plots. Yellow lines show the cut values that create the containment volume, with arrows pointing towards the preserved events.

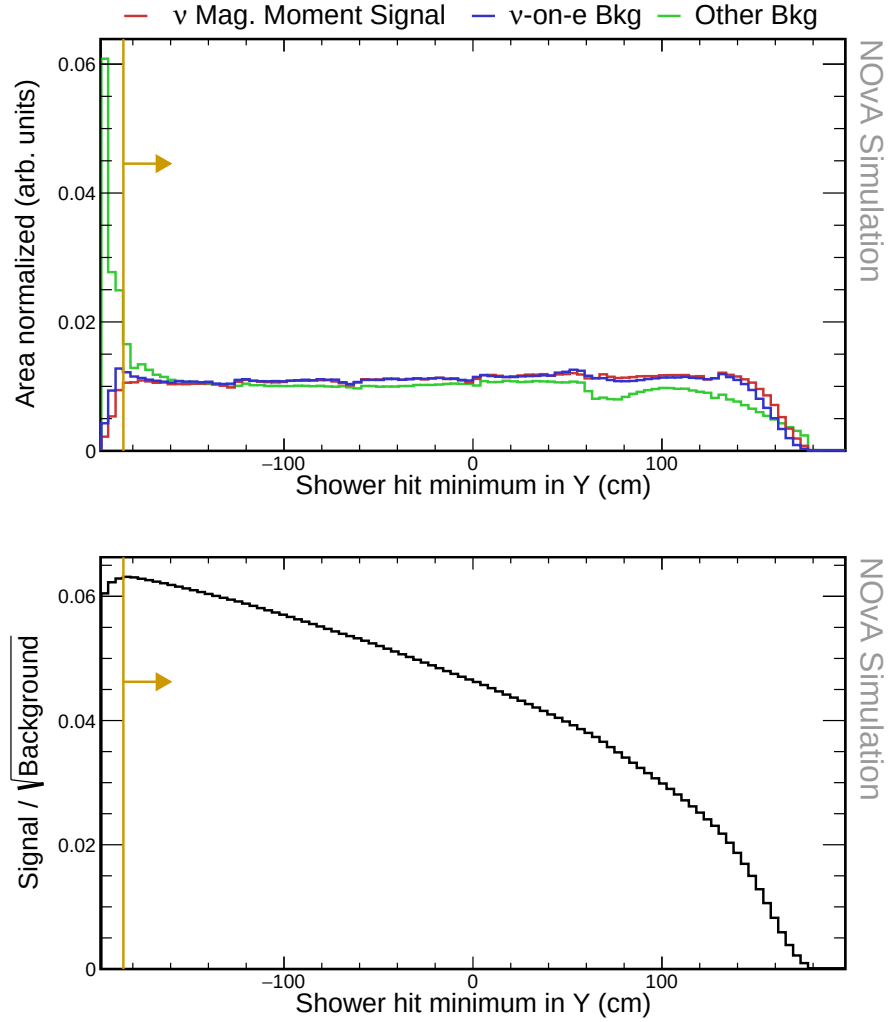


Figure 17: Top: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the minimum hit position of the most energetic prong along the y axis. All histograms are area-normalized. Bottom: Cumulative FOM calculated as the number of signal events, divided by the number of background events from that bin until the end of the plot in the direction of the yellow arrow. The reconstruction quality, pre-selection and fiducial cuts were applied prior to making these plots. Yellow lines show the cut values that create the containment volume, with arrows pointing towards the preserved events.

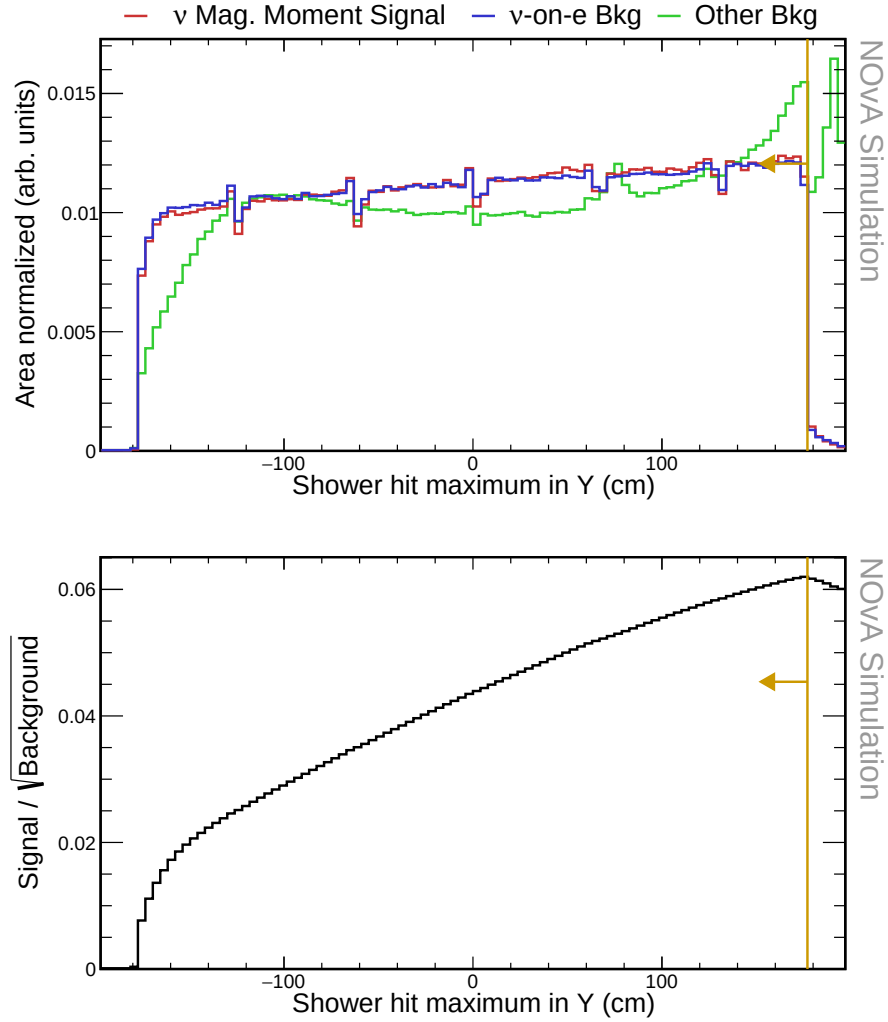


Figure 18: Top: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the maximum hit position of the most energetic prong along the y axis. All histograms are area-normalized. Bottom: Cumulative FOM calculated as the number of signal events, divided by the number of background events from that bin until the end of the plot in the direction of the yellow arrow. The reconstruction quality, pre-selection and fiducial cuts were applied prior to making these plots. Yellow lines show the cut values that create the containment volume, with arrows pointing towards the preserved events.

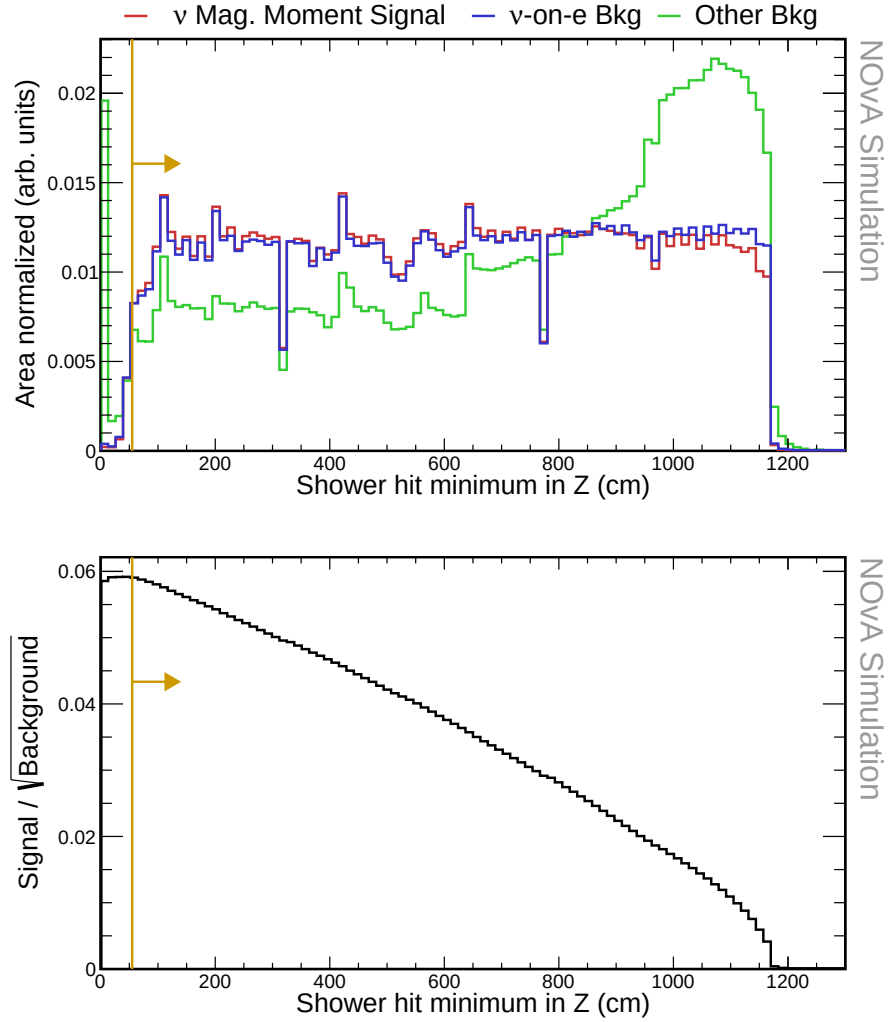


Figure 19: Top: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the minimum hit position of the most energetic prong along the z axis. All histograms are area-normalized. Bottom: Cumulative FOM calculated as the number of signal events, divided by the number of background events from that bin until the end of the plot in the direction of the yellow arrow. The reconstruction quality, pre-selection and fiducial cuts were applied prior to making these plots. Yellow lines show the cut values that create the containment volume, with arrows pointing towards the preserved events.

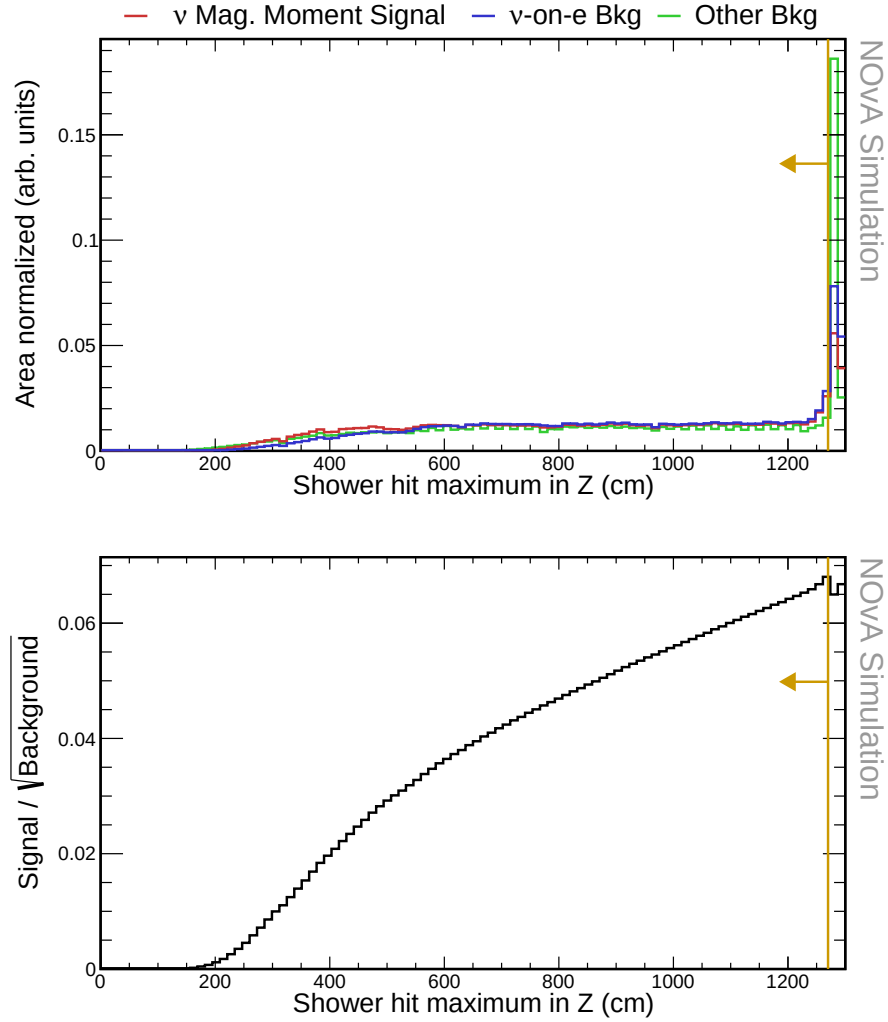


Figure 20: Top: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the maximum hit position of the most energetic prong along the z axis. All histograms are area-normalized. Bottom: Cumulative FOM calculated as the number of signal events, divided by the number of background events from that bin until the end of the plot in the direction of the yellow arrow. The reconstruction quality, pre-selection and fiducial cuts were applied prior to making these plots. Yellow lines show the cut values that create the containment volume, with arrows pointing towards the preserved events.

magnetic moment. This goal is equivalent to maximising our FOM from Eq. 39.

For this purpose, we utilised ROOT's [34] Tool for MVA (TMVA) [35]. Specifically, we employed the rectangular cut optimisation method, which uses multivariate parameter fitters to maximise background rejection across the full range of signal efficiencies. We used the MC sampling fitting method, assuming that for each input variable, there is a single cut value (maximum or minimum) that optimally discriminates between signal and background.

TMVA generally performs better with a limited number of input variables that have strong discriminating power. Therefore, we investigated several input variables and selected only those that achieved significant background rejection. There are additional variables not mentioned here that might achieve better final results, providing opportunities for future re-analyses. Additionally, we do not apply any transformations to the input variables prior to optimisation, which might also improve the final result after dedicated study.

The variables considered include those already used in the pre-selection: the total number of hits for all prongs in slice, the length of the longest prong, and $E\theta^2$, as discussed in Sec. 4.3. During the TMVA optimisation, we found that the length of the longest prong did not significantly enhance discriminating power and thus removed it from the set of input variables. Additionally, we included the reconstructed energy of the most energetic shower, as used for reconstruction quality selection in Sec. 4.2, intending to restrict events with higher energies, since our signal is concentrated at low electron recoil energies.

Additionally, we considered all the variables used for the NOvA ν -on-e analysis for the neutrino flux constraint [13]. The first is the fraction of the reconstructed energy of the most energetic shower (E_{Shower}) to the total energy of all the reconstructed prongs in the entire slice (E_{Tot}). This variable distinguished our signal events, which only have a single shower, from events with multiple showers or additional activity. The second is the gap between the vertex and the most energetic shower, which can distinguish between electron and π^0 events, as the latter should have a characteristic gap several cells long. Additionally, we examined the amount of energy contained within ± 8 planes away from the vertex, besides the energy associated with the most energetic prong, which should distinguish the purely leptonic signal from backgrounds with significant hadronic activity. However, the gap and the vertex energy variables underperformed compared to others and were ultimately not used within the TMVA.

We also utilised two Convolutional Neural Network (CNN)-based event classifiers developed by the ND group for the NOvA ν -on-e analysis for the neutrino flux constraint [13]. These classifiers are specifically designed to identify ν -on-e interactions. The first, named ν -on-eID, is trained to select ν -on-e events from the primary ν_μ CC background, while the second, named $E\pi^0$ ID, is trained on events passing the ν -on-eID selection to reject the remaining background with a π^0 . These classifiers use a pixel map of the entire slice as input and are designed with the same CNN architecture as ProngCVN and EventCVN.

The result of the TMVA is a set of cuts on each of the input variables that maximises the FOM. The input variables and the cuts that were selected for them are shown in Fig. 21, 22, and 23. The effect of these cuts is summarised in Tab. 9. Applying the TMVA cuts reduces the signal by 51.62 %, the ν -on-e background by 75.03 % and other background by 99.98 %. The specific values of the cuts resulting from

the TMVA are

$$E_{\text{Shower}}/E_{\text{Tot}} > 0.91, \quad (46)$$

$$\text{Total } N^o \text{ hits for all prongs} < 116, \quad (47)$$

$$E_{\text{Shower}} < 1.4 \text{ GeV}, \quad (48)$$

$$E\theta^2 < 0.0048 \text{ GeV} \times \text{rad}^2, \quad (49)$$

$$\nu - \text{on} - \text{eID} > 0.65, \quad (50)$$

$$E\pi^0\text{ID} > 0.63. \quad (51)$$

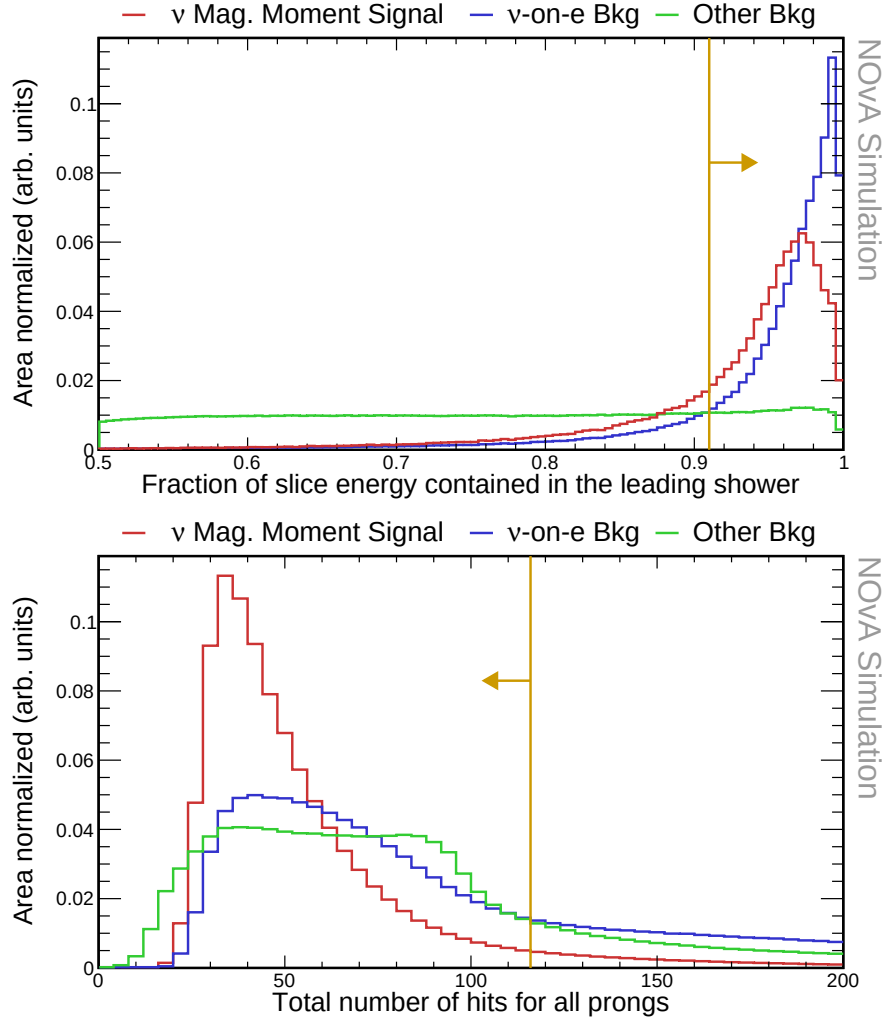


Figure 21: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the fraction of the total energy contained in the primary shower (top) and of the total number of hits in the slice (bottom). All histograms are area-normalized. The reconstruction quality, pre-selection, fiducial and containment cuts were applied prior to making these plots. Yellow lines show the cut values on the depicted variables, with arrows pointing towards the preserved events.

After the full event selection, the predicted number of signal events for $\mu_\nu = 10^{-9}\mu_B$ is 56.80, and the total number of background events under the SM hypothesis is 700.33. The decomposition of

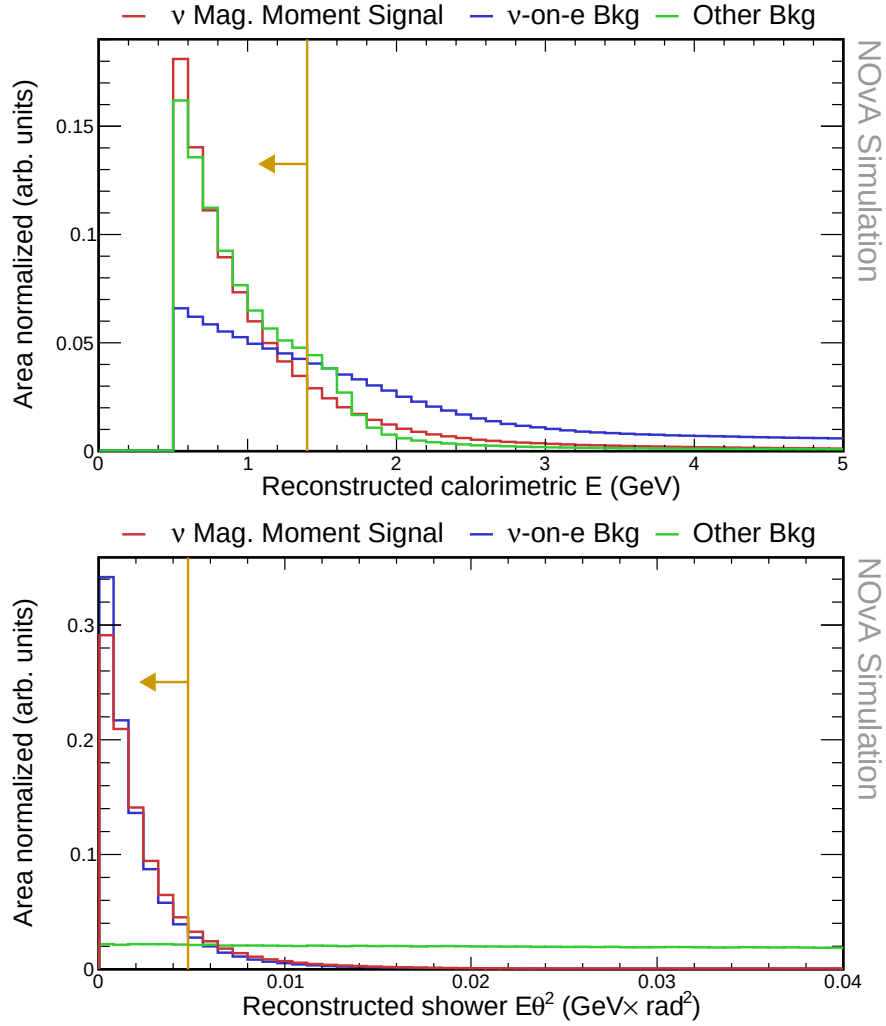


Figure 22: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the reconstructed energy of the primary shower (top) and of the reconstructed energy multiplied by the angle from the incoming neutrino beam direction squared.(bottom). All histograms are area-normalized. The reconstruction quality, pre-selection, fiducial and containment cuts were applied prior to making these plots. Yellow lines show the cut values on the depicted variables, with arrows pointing towards the preserved events.

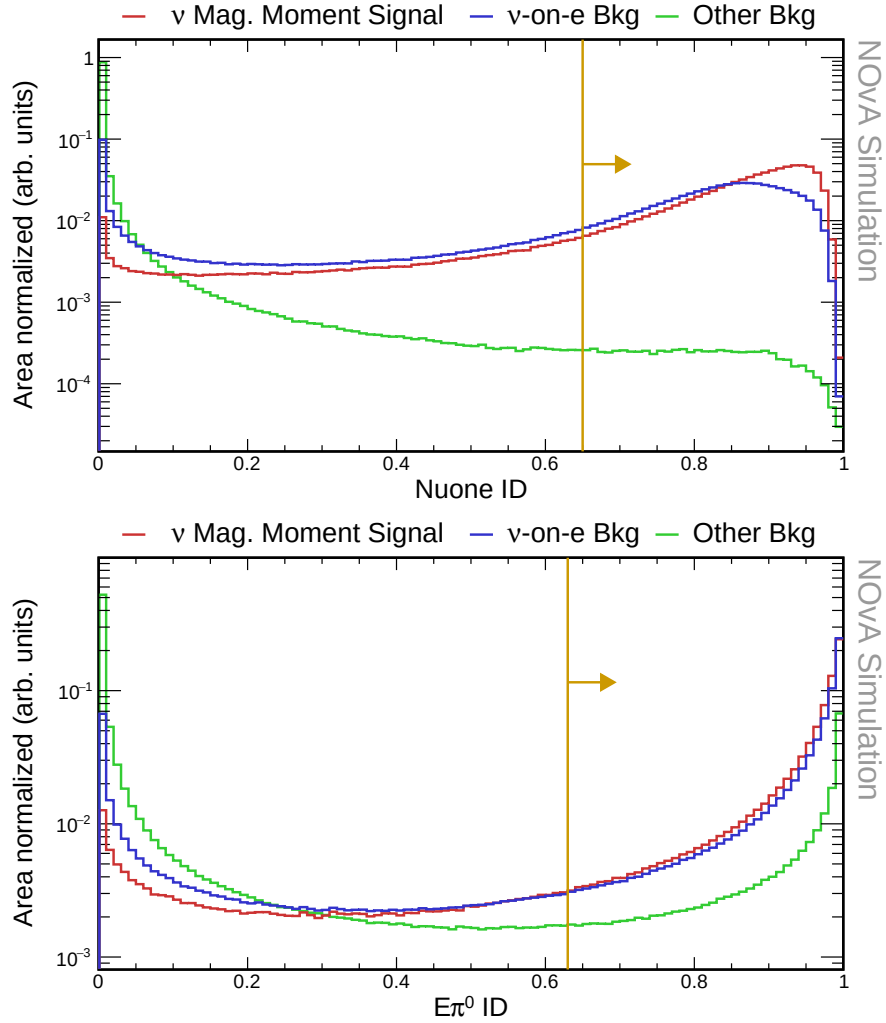


Figure 23: Relative comparison of signal (red), ν -on-e background (blue), and other background (green) events in the distribution of the NuoneID (top) and EPi0ID (bottom) event identifiers. All histograms are area-normalized and logarithmic in the y axis. The reconstruction quality, pre-selection, fiducial and containment cuts were applied prior to making these plots. Yellow lines show the cut values on the depicted variables, with arrows pointing towards the preserved events.

background into interaction types is shown in Fig. 24 and listed in Tab. 10. The event selection results in

$$\text{Signal Purity} = \frac{\text{Signal}}{\text{Signal} + \text{Background}} = 7.50\%, \quad (52)$$

$$\text{Signal Efficiency} = \frac{\text{Signal}}{\text{Signal}_{\text{No Cut}}} = 6.95\%. \quad (53)$$

Also,

$$\frac{\text{Signal}}{\sqrt{\text{Background}}} = 2.15 \quad (54)$$

and

$$\text{FOM} = \frac{\text{Signal}}{\sqrt{\text{Signal} + \text{Background}}} = 2.06. \quad (55)$$

Table 9: Event selection cutflow table for the results of the cut-based Multivariate analysis, showing the number of events and the relative efficiency of each cut for each signal sample. The relative efficiency is calculated as number of events remaining after applying the corresponding cut divided by number of event for all the previous cuts. All the cuts are listed in sequence as they are applied. The top row corresponds to the sample after applying the reconstruction quality, pre-selection, fiducial and containment cuts.

Selection	Signal		ν -on-e bkg		Other bkg	
	N_{evt}	$\epsilon_{\text{rel}}(\%)$	N_{evt}	$\epsilon_{\text{rel}}(\%)$	N_{evt}	$\epsilon_{\text{rel}}(\%)$
Contained	117.41	100	2.08×10^3	100	1.10×10^6	100
$E_{\text{Shower}}/E_{\text{Tot}}$	113.03	96.28	2.02×10^3	97.32	4.53×10^5	41.30
N^o Hits	106.48	94.20	1.45×10^3	71.53	4.02×10^5	88.78
High E_{Shower}	85.51	80.31	777.91	53.76	3.01×10^5	74.84
ν-on-e ID	72.23	84.47	652.32	83.86	4.40×10^3	1.46
$E\pi^0$ ID	67.35	93.24	608.19	93.23	2.83×10^3	64.34
$E\theta^2$	56.80	84.33	519.09	85.35	181.24	6.40

Table 10: Interaction types contributing to the SM background after full event selection. Interactions with/without π^0 in the final state are specifically selected due to their significant contribution to the ν -on-e background.

Interaction	Number of events
ν -on-e	519.09
NC w/ π^0	72.96
NC w/o π^0	21.51
ν_μ CC w/ π^0	28.07
ν_μ CC w/o π^0	25.67
ν_e CCMEC	2.22
Other ν_e CC	9.83
Other	20.98
Total	700.33

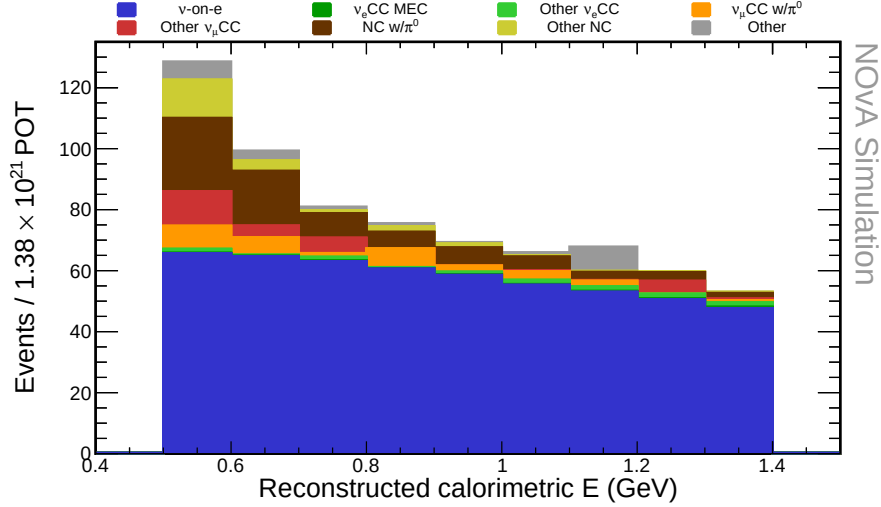


Figure 24: Background composition of events passing the full event selection as a function of the reconstructed energy of the leading shower - electron for ν -on-e events. Events are scaled to the data exposure. Origin of the "Other" background events between 1.1 – 1.2 GeV is not known.

5 Systematic Uncertainties

We consider all the standard NOvA systematic uncertainties grouped into four categories: neutrino flux, detector calibration, detector modelling, and neutrino interaction systematic uncertainties. Summary of the effects of both systematic and statistical uncertainties on the predicted number of SM background events is shown in Fig. 25. The four categories of systematic uncertainties are assumed to be uncorrelated between each other, allowing to calculate their combined effect by adding their individual contribution in quadrature. The statistical uncertainty is calculated as $\sqrt{N_{SM}}$ since the number of predicted events follows the Poisson distribution. The total prediction of the number of SM background events can be expressed as

$$N_{SM} = 700.33 \pm 26.46 \text{ (stat.) }^{+72.48}_{-62.99} \text{ (syst.)}. \quad (56)$$

To assess the effect of the neutrino flux systematic uncertainty, we use 8 ND-only principal components. Figure 26 shows their combined effect on the predicted SM background as a function of the primary shower's reconstructed energy. Since the principal components are uncorrelated by construction, the total systematic uncertainty for each bin is calculated by adding the effect of all the principal components in quadrature. The effect of the neutrino flux systematic uncertainty does not depend on the primary shower's calorimetric energy and altering the neutrino flux prediction can be represented by a normalization shift. The final effect of the neutrino flux systematic uncertainty on the total number of SM background events is $\pm 8.16\%$ and is symmetric around the nominal prediction.

The effect of the total detector modelling and calibration systematic uncertainties on the number of SM background events is asymmetrical, increasing the number of events by $+6.17\%$ and decreasing by 3.69% . The effect of these uncertainties depends on the energy of the primary shower (which is the recoil electron for ν -on-e events), as can be seen in Fig. 27. This means that the energy distribution of the SM background events can be significantly altered due to the detector uncertainties. The largest contribution comes from the absolute energy scale uncertainty, which has a one-sided effect of $+5.72\%$. This is due to both the positive and the negative shift in the absolute energy scale increasing the total number of SM background events. The light level and Cherenkov systematic uncertainties are asymmetrical and alter the number of SM background events in both directions by $^{+1.13\%}_{-3.39\%}$ and $^{+1.82\%}_{-1.46\%}$ respectively. The

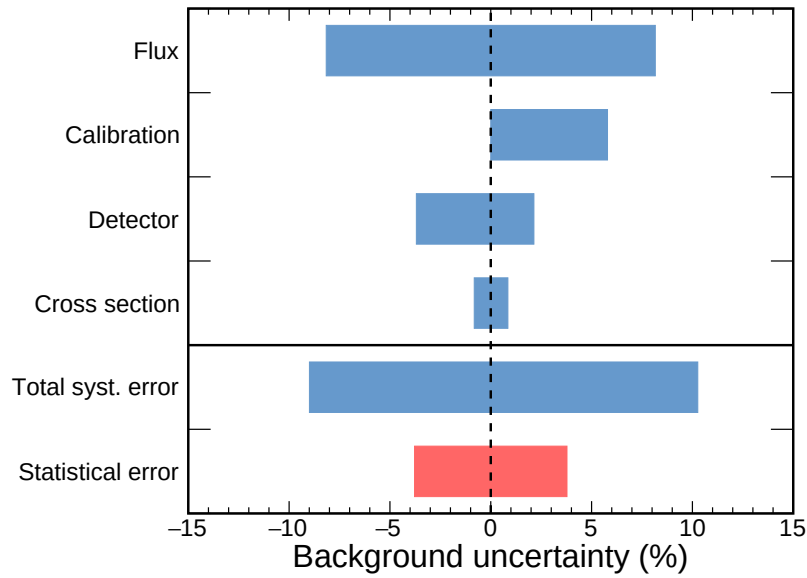


Figure 25: Relative effect of systematic and statistical uncertainties on the number of SM background events. The total systematic uncertainty (blue bottom bar) is calculated as a square root of the sum of squares of the four categories of systematic uncertainties, shown ordered by the size of their effect. These are the neutrino flux, detector calibration, detector modelling, and neutrino cross section systematic uncertainties.

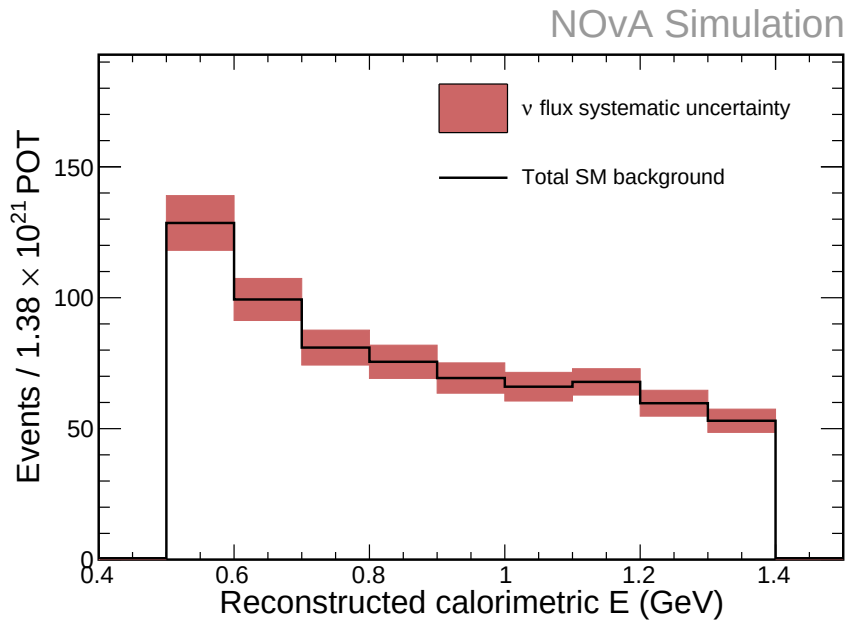


Figure 26: Effect of the total neutrino flux systematic uncertainty on the primary shower energy distribution of the SM background events.

detector ageing and calibration shape systematic uncertainties are one sided by default and increase the number of the SM background events by +0.55 % and +0.69 % respectively.

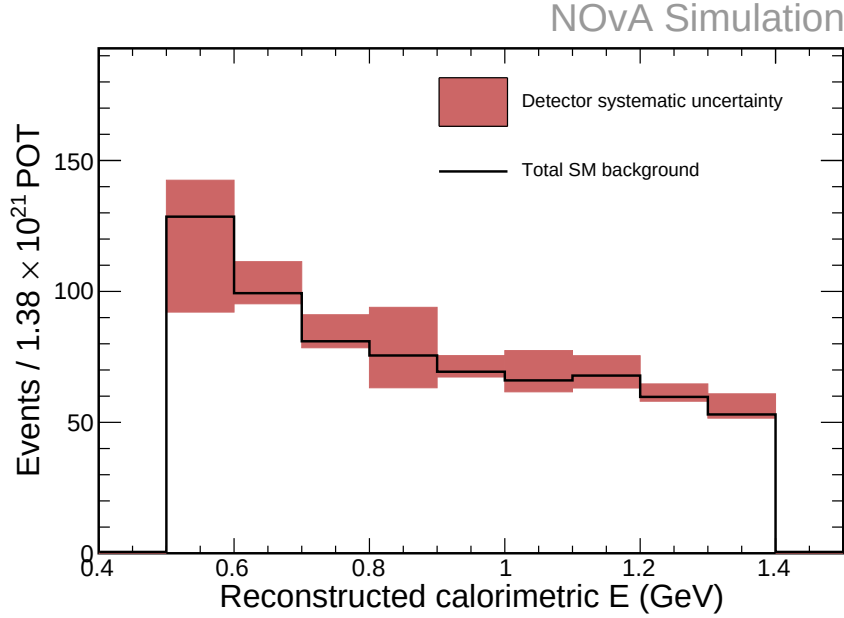


Figure 27: Effect of the total detector systematic uncertainty on the primary shower energy distribution of the SM background. The detector uncertainty consists of the absolute calibration, calibration shape, detector ageing, light level, and Cherenkov systematic uncertainties.

Since we assume that the ν -on-e interaction is known precisely, the dominant ν -on-e component of the SM background has no systematic uncertainty from neutrino interactions. Therefore, the neutrino interaction systematic uncertainty affects only the 181 "Other background" events, which make up 34.9 % of the total SM background. Consequently, the total neutrino interaction uncertainty on the SM background is relatively small, with a combined effect on the number of SM background events of $+1.54\%$ $_{-1.49\%}^{+1.54\%}$. The most dominant neutrino interaction systematic uncertainties are related to interactions with π^0 in the final state, such as the axial mass ($+0.97\%$ $_{-0.90\%}^{+0.97\%}$) and the vector mass ($+0.41\%$ $_{-0.35\%}^{+0.41\%}$) of the NCResonant baryon production (Res) interactions, the scaling of the Coherent π (COH π)NC interactions ($+0.85\%$ $_{-0.85\%}^{+0.85\%}$), or the mean free path of pions before they undergo an interaction ($+0.30\%$ $_{-0.43\%}^{+0.30\%}$). Other neutrino interaction systematic uncertainties have only a marginal ($< 0.3\%$) effect on the number of SM background events.

6 Statistical Analysis

6.1 Resolution and binning

(TO DO: If there is going to be a fit to the electron recoil energy there probably need to be some studies done into the electron energy and angle resolution and binning)

6.2 Counting Experiment

Our analysis compares the total observed number of events, N_O , with the expected (predicted) number of events, $N_E(\mu_\nu)$, which in general depends on the value of the effective neutrino magnetic moment,

μ_v and is the sum of the expected signal events, $N_S(\mu_v)$, and the expected background events, N_B . The number of recorded events within a fixed time interval follows the Poisson distribution with a likelihood (probability) of observing N_O events, given a hypothesis predicting N_E events, is defined as:

$$L(N_O; N_E(\mu_v)) = \frac{N_E(\mu_v)^{N_O}}{N_O!} e^{-N_E(\mu_v)}. \quad (57)$$

To estimate and constrain μ_v , we employ a frequentist approach using the Poisson-distributed likelihood ratio [36]

$$\lambda = \frac{L(N_O; N_E(\mu_v))}{L_{sat}(N_O; N_O)}. \quad (58)$$

Here, L_{sat} is the so-called saturated model, for which $N_E = N_O$ [19]. Maximising the likelihood with respect to μ_v is equivalent to minimising the ‘log-likelihood ratio’ described as:

$$-2 \ln \lambda(\mu_v) = -2 \ln \left(\frac{L(N_O; N_E(\mu_v))}{L(N_O; N_O)} \right) \quad (59)$$

$$= 2 \left[N_E(\mu_v) - N_O + N_O \ln \frac{N_O}{N_E(\mu_v)} \right]. \quad (60)$$

According to Wilk’s theorem [37], under the assumptions that the hypothesis is correct, the sample size is sufficiently large, and certain regularity conditions (which are met in our case) are satisfied, the log-likelihood ratio follows a chi-squared (χ^2) distribution. Therefore, the log-likelihood ratio serves not only to find the maximum likelihood estimators of μ_v but also to provide a statistic usable for a goodness-of-fit test [19].

6.2.1 Treatment of Systematic Uncertainties

Systematic uncertainties are incorporated into the fitting framework as nuisance parameters that multiply the likelihood function in Eq. 57. Each systematic uncertainty is modelled according to the normal distribution centred at zero, with a standard deviation corresponding to its respective 1σ variation. Systematic uncertainties are treated as mutually uncorrelated, justifying their independent treatment.

Incorporating the nuisance parameters into the log-likelihood ratio function in Eq. 59 results in so-called ‘penalty terms’. These terms are proportional to the square of each systematic pull’s variation in units of σ , thereby increasing the log-likelihood ratio for solutions that favour non-zero systematic shifts:

$$-2 \ln \lambda(\mu_v, \vec{\eta}) = -2 \left[E(\mu_v, \vec{\eta}) - N_O + N_O \ln \frac{N_O}{E(\mu_v, \vec{\eta})} \right] + \sum_{i=1}^M \left(\frac{\eta_i}{\sigma} \right)^2. \quad (61)$$

Here, $\vec{\eta}$ is a vector of M nuisance parameters representing the effect of systematic uncertainties.

In practice, the systematic variations are initially predicted at fixed integer σ values. Due to the computationally expensive nature of reproducing the detector simulation, variations in detector and calibration systematic uncertainties are estimated only at $\pm 1\sigma$ values. However, for beam and cross section uncertainties, variations are estimated at $\pm 1\sigma$, $\pm 2\sigma$, and $\pm 3\sigma$ levels. These systematically shifted predictions are then interpolated using of cubic spline-based fit, which is fixed at zero for the excluded sides of one-sided systematic uncertainties, such as detector ageing and calibration shape [38].

6.2.2 Hypothesis Testing and Confidence Limits

To evaluate whether the observed number of events can be described solely using the SM prediction, we test the ‘null hypothesis’, for which $\mu_\nu = 0$ and therefore the expected number of events is $N_E = N_B$. The log-likelihood ratio is in this case

$$-2 \ln \lambda(\theta) = 2 \left[N_B - N_O + N_O \ln \frac{N_O}{N_B} \right] + \sum_{k=1}^M \eta_k^2. \quad (62)$$

The minimum of this log-likelihood ratio follows a χ^2 distribution with one degree of freedom, which allows us to calculate the p-value of the goodness-of-fit of the null hypothesis. We reject the null hypothesis if the statistical test yields a p-value below a critical value - significance level. We set the significance level at $\alpha = 0.10$, which is a common threshold in other measurements of the neutrino magnetic moment, as discussed in the introduction to this chapter.

If we fail to reject the null hypothesis ($p > \alpha$), we establish an upper limit on the effective muon neutrino magnetic moment. We consider our results to represent only the ν_μ thanks their high purity in our neutrino beam. The upper limit is calculated by iterating over a fixed range of neutrino magnetic moment values and profiling over the systematic shift variations. This means that at each iteration, the value of the neutrino magnetic moment is fixed, and the log-likelihood function is minimised only with respect to the systematic shifts. The minimisation of the log-likelihood ratio in Eq. 61 is done using a MIGRAD-based algorithm in ROOT [34]. The minimum of the log-likelihood, as a function of the neutrino magnetic moment, follows a χ^2 distribution with one degree of freedom. Exclusion regions and limits are then derived based on integrating the tail of the χ^2 distribution tables with one degree of freedom.

6.3 Fitting framework

(TO DO: Describe the fitting framework, ideally with an example plot, maybe with some arrows) How does the fitting framework work? It’s based on the framework developed by Mu Wei for the Light Dark Matter analysis (ref.) which was developed together (is this fair?). Basic description of the framework.

Also this framework is used for both LDM and NuMM together. It is trivial to simply switch between including the NuMM or LDM in it. This was done to save space in creating predictions since our backgrounds are exactly the same (or at least they should be...). Theoretically this could be separated into two difference frameworks.

- <NDPredictionSingleElectron> Prediction class which holds the LDM as a special 2-D spectrum (not used for NuMM), and NuMM, ν -on-e background, ν_e CC MEC background and other background as simple 1-D spectra. Also scaling each spectra by...
- <NDPredictionSystSingleElectron> class derived from PredictionInterp that takes in the NDPredictionSingleElectron and applies systematic shifts to it. Includes the interpolation/extrapolation between the systematic shifts.
- FitVariables and what do they do
- Fitter which does exactly what... What are the parameters of the fit? What are the results/outputs?

7 Results / Fake data studies

[to do]: talk about the shape-only versus normalisation-only strategies for the differential versus single-bin analyses.

I think it might be a good idea to talk about this already in the analysis overview section.

I need to also discuss fake data studies. Should I talk about this now? Probably

7.1 Counting experiment

Single bin analysis - total numbers of signal and backgrounds predicted. Possible background corrections and its effect on the background uncertainties.

7.2 Binned experiment

Plots showing delta chi2 for stats and systs separately. Plot showing the chi2 with limits. Maybe talk about using different binning and variables and the effect.

Also need to somehow quantify the various fitter settings, like different seeding values, different range of profiling. Should I talk about Feldman-Cousins?

7.3 Template fit

(TO DO: *Describe the fitting framework, ideally with an example plot, maybe with some arrows*) How does the fitting framework work? It's based on the framework developed by Mu Wei for the Light Dark Matter analysis (ref.) which was developed together (is this fair?). Basic description of the framework.

Also this framework is used for both LDM and NuMM together. It is trivial to simply switch between including the NuMM or LDM in it. This was done to save space in creating predictions since our backgrounds are exactly the same (or at least they should be...). Theoretically this could be separated into two difference frameworks.

- `<NDPredictionSingleElectron>` Prediction class which holds the LDM as a special 2-D spectrum (not used for NuMM), and NuMM, ν -on-e background, ν_e CC MEC background and other background as simple 1-D spectra. Also scaling each spectra by...
- `<NDPredictionSystSingleElectron>` class derived from `PredictionInterp` that takes in the `NDPredictionSingleElectron` and applies systematic shifts to it. Includes the interpolation/extrapolation between the systematic shifts.
- `FitVariables` and what do they do
- Fitter which does exactly what... What are the parameters of the fit? What are the results/outputs?

7.3.1 Sensitivities and limits

8 Conclusion

(TO DO: *Report the limit with its uncertainty*)

(TO DO: *Very briefly discuss differences with current world limit and how the techniques differ*)

533 (TO DO: *Very briefly summarise expectations for future measurements.*)

534 Acronyms

535 **ν -on-e** neutrino-on-electron (interaction). 1, 3, 5–10, 12, 13, 15–24, 26–40, 42

536 **APD** Avalanche Photodiode. 17

537 **BSM** Beyond Standard Model. 1–4

538 **C.L.** Confidence Level. 1

539 **CC** Charged Current. 7, 13–15, 21, 35, 39

540 **CNN** Convolutional Neural Network. 35

541 **COH π** Coherent π (production). 42

542 **CVN** Convolutional Visual Network. 16, 17

543 **DCM** Data Concentration Module. 17

544 **FEB** Front End Board. 17

545 **FOM** Figure Of Merit. 15, 21, 26–35, 39

546 **LDM** Light Dark Matter. 1, 10, 15

547 **MC** Monte Carlo. 10, 12, 35

548 **MEC** Meson Exchange Current. 13–15, 39

549 **ML** Machine Learning. 17

550 **MVA** Multi Variate Analysis. 15, 21, 35, 47

551 **NC** Neutral Current. 39, 42

552 **ND** Near Detector. 1, 3, 5, 9–11, 15, 21, 35, 40

553 **NO ν A** NuMI Off-axis ν_e Appearance (experiment). 1, 3, 5–11, 13, 15, 17, 21, 35, 40

554 **NP** New Physics. 4

555 **NuMI** Neutrinos from the Main Injector. 25

556 **PMNS** Pontecorvo-Maki-Nakagawa-Sakata. 3, 4

557 **POT** Protons On Target. 10, 11, 13, 17, 20

558 **PPFX** Package to Predict the Flux. 12

559 **Res** Resonant baryon production. 42
 560 **SM** Standard Model. 1–4, 7–10, 12, 16, 36, 39–42, 44
 561 **TMVA** Tool for MVA. 35, 36

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